

Autonomic Nervous System Pharmacology

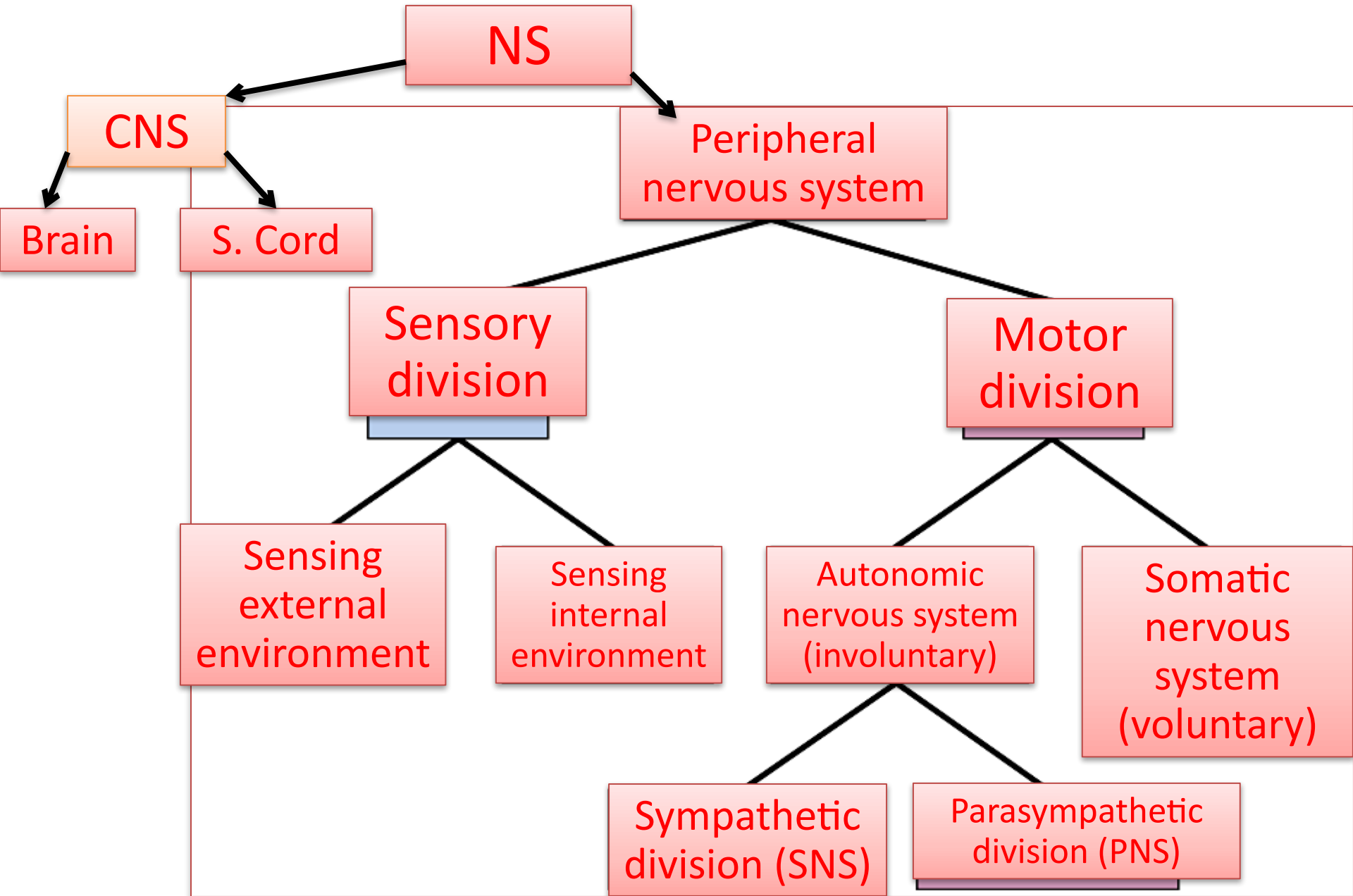
Objectives

- ❖ After completing this chapter the students able to:
 - ❑ Understand Neuronal transmission in ANS
 - ❑ Know the Effect of Cholinergic and Adrenergic innervations in Different organ-systems
 - ❑ Understand MOA, indications, contraindications, and Adverse effect of Cholinergic and anti-cholinergic Drugs
 - ❑ Understand MOA, Indications, Contraindications, Adverse effect of Adrenergic and anti-adrenergic drugs

Outlines

- ❑ Terminologies
- ❑ Introduction
- ❑ Neurotransmission
 - Steps of transmission
 1. Axonal conduction
 2. Synaptic neurotransmission
 - Cholinergic drugs
 - Adrenergic drugs
 - Anti-cholinergic Drugs
 - Anti-Adrenergic drugs

Introduction



Introduction-----

- ❖ The autonomic system (ANS) regulates the vital body functions **without the conscious participation** of the mind.
- ❖ The ANS is composed of efferent neurons that innervates smooth muscle of :
 - the viscera - cardiac muscle,
 - vasculature - the exocrine glands,
- ❖ There by controlling **digestion, cardiac output, blood flow**, and **glandular secretions**.
- ❖ Autonomic drugs act either by **stimulating** or by **blocking** the action of the autonomic nerves.

Introduction-----

Sympathetic Vs Parasympathetic

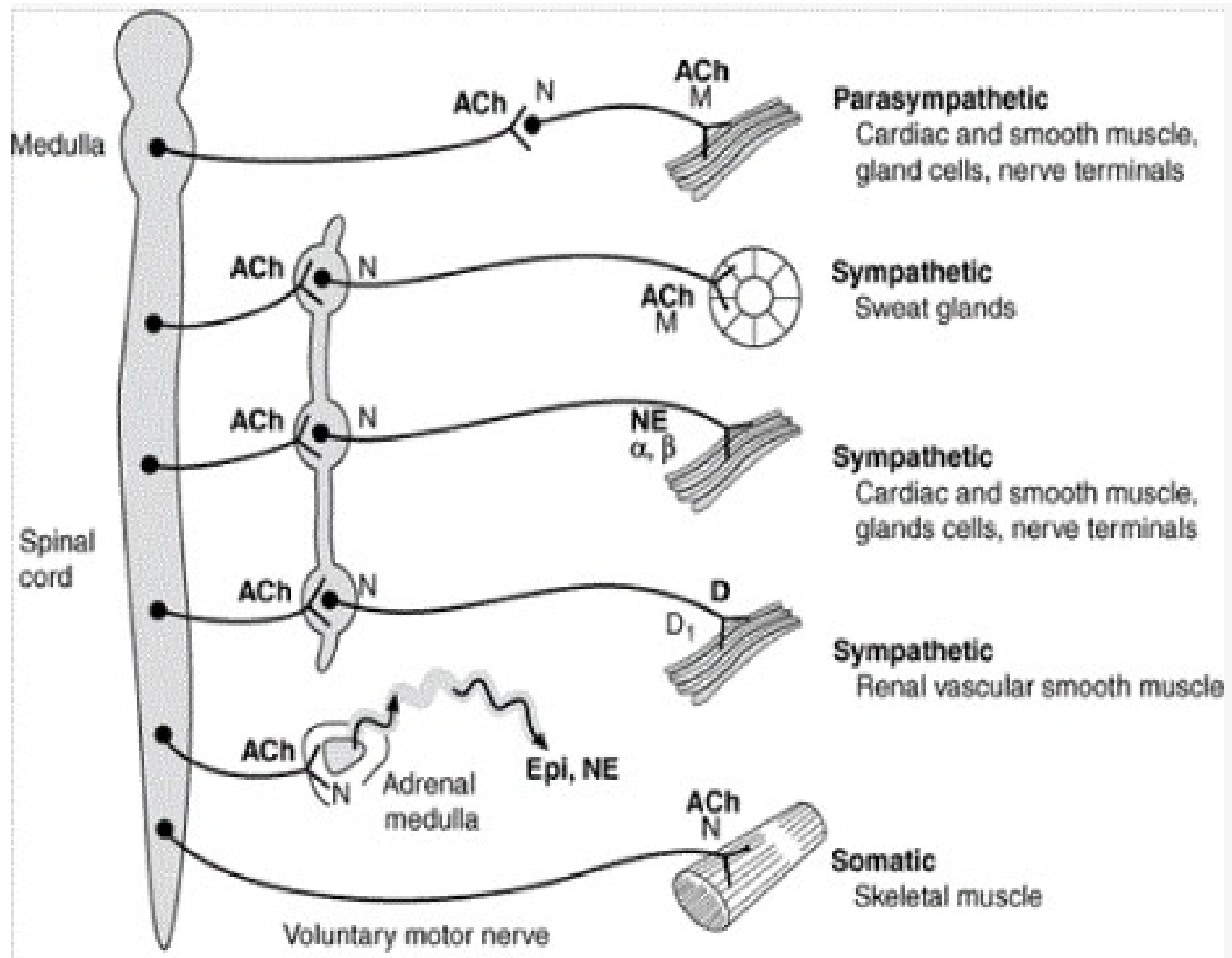
❖ **Peripheral nervous system provides** a double set of nerve fibers:

❑ **Sympathetic (adrenergic)**

- Fight or flight action
- Exit from **thoracic and lumbar regions**
- Neurotransmitter- **norepinephrine**
- Preganglionic neuron is **shorter** than postganglionic neuron
- Adrenergic receptors- $\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$, $\beta 3$

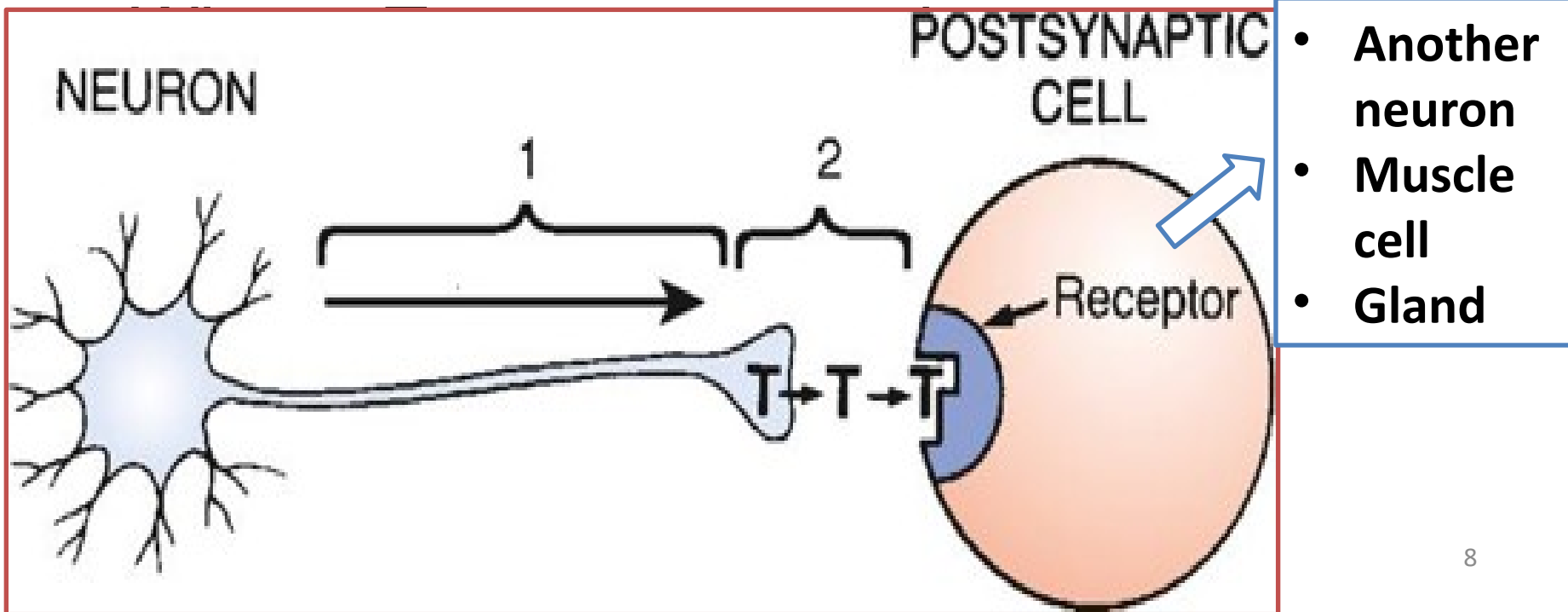
❑ **Parasympathetic (cholinergic)**

- Rest or digest
- Exit from **cranial and sacral portions**
- Neurotransmitter- **acetylcholine**
- Preganglionic neuron is **longer** than postganglionic neuron



Neurotransmission

- There are two basic steps in the process by which signal is transmitted b/n a neuron and a postsynaptic cell:
 1. Axonal conduction (electrical signal)
 2. Synaptic transmission (NT/chemical signal)



Neurotransmission -----

❖ Axonal conduction

- The **process of conducting an action potential** down the axon of the neuron
- Not common site of drugs' action
 - Because they are non-selective
 - As impulse conduction along all axon is the same
- Local anesthetics are the only drugs proved to work by altering (decreasing) axonal conduction.

Neurotransmission -----

❖ Synaptic transmission

- The **process by which information is carried across the gap between the neuron and the postsynaptic cell.**
- Requires the **release of NT molecules** from the axon terminal
- Followed by binding of these molecules to **receptors on the postsynaptic cell.**
- drugs that alter synaptic transmission can produce effects that are highly **selective.**
- Why?
 - Because synapses, unlike axons, differ from one another.
 - Synapses at different sites employ different transmitters.
 - Most transmitters have more than one type of receptor.
- Thus, it is main site of drugs' action

❖ **Major neurotransmitters include:**

- **Acetylcholine**
- **Catecholamine's (Nor epinephrine, Epinephrine)**

❖ **Acetylcholine** is the transmitter released by:

- **All preganglionic neurons of the PNS,**
- **All postganglionic neurons of the PNS,**
- All motor neurons to **skeletal muscles**
- **Postganglionic neurons of the SNS that go to sweat glands**

❖ **Norepinephrine** is the transmitter released by:

- Practically all **postganglionic neurons** of the SNS, except those run to **sweat gland**.

❖ **Epinephrine** is the major transmitter released by the adrenal medulla

Neurotransmission -----

❖ Steps in Synaptic Transmission

1. Synthesis of transmitter from precursor molecules
2. Storage of transmitter in vesicles.
3. Release of transmitter
4. Action at (binding to) receptor
5. Termination of transmission by:
 - Reuptake into the presynaptic nerve terminal
 - Enzymatic dégradation
 - ✓ Acétylcholine estérase ,for Ach
 - ✓ Monoamine oxidase/MAO for catéchol amines
 - Diffusion away from junction
 - Uptake by extra-neuronal sites

Cholinergic neurotransmission

Ach synthesis, storage and transport

- Rate limiting step is choline transported by a sodium-dependent choline transporter (CHT).
- Ach is synthesized
 - from choline and acetyl Co-A (AcCoA)
 - by the enzyme choline acetyltransferase (ChAT).

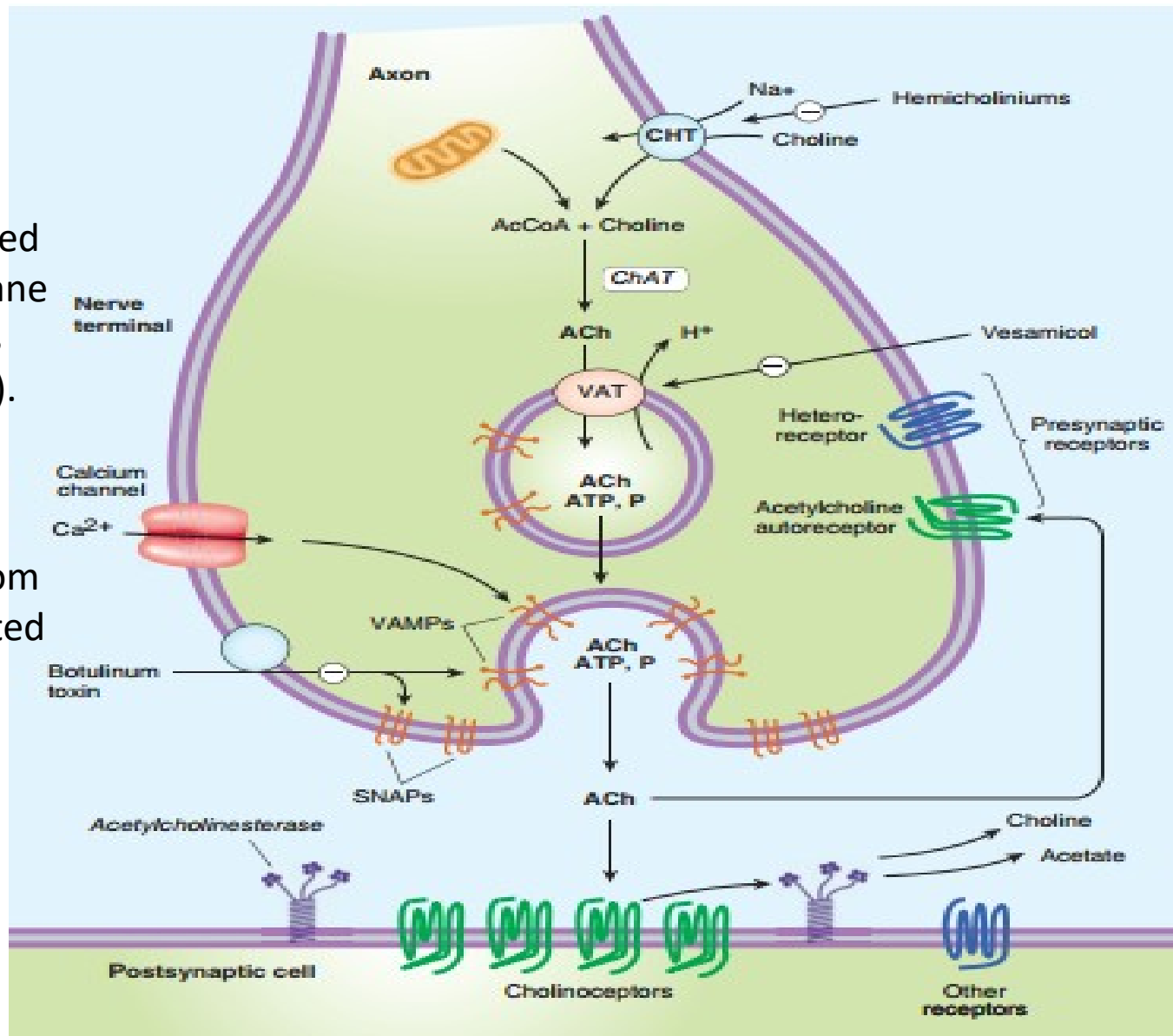
- ACh is then transported into the storage vesicle
 - by a carrier, the vesicle-associated transporter (VAT)
- **Release** occurs when
 - voltage-sensitive calcium channels in the terminal membrane are opened → entry of calcium → increase in intracellular calcium → fusion of vesicles with the surface membrane → exocytosis → Ach interacts with cholinoreceptors.

- Acetylcholine's action is terminated
 - Metabolism by acetylcholinesterase (AChE)
 - Highly concentrated on the outer surfaces of both the prejunctional (neuronal) and postjunctional (effector cell) membranes.
- Short life span - A rapid hydrolysis of acetylcholine
- AChE is also found in other tissues, eg, RBC
- Butyrylcholinesterase
 - Another cholinesterase with a lower specificity for ACh

- Pseudocholinesterase
 - Found in blood plasma, liver and many other tissues
- Little or no acetylcholine escapes into the circulation
 - inactivated by plasma esterases.
- Receptors on the presynaptic nerve ending regulate transmitter release

vesicle-associated membrane proteins (VAMPs).

synaptosome-associated proteins



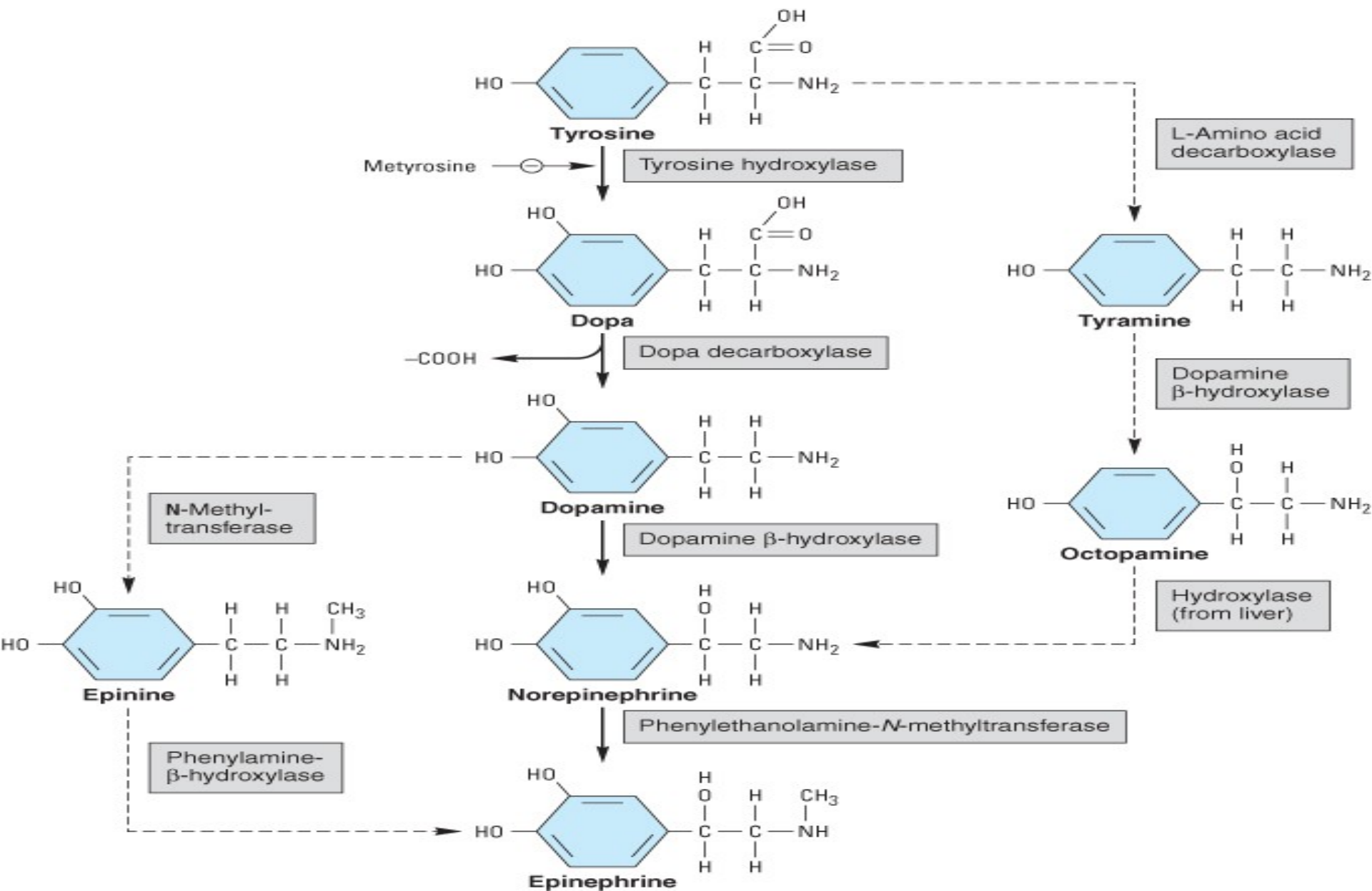
Adrenergic Transmission

- **Synthesis of NA**

- Tyrosine is converted to dopamine → transported into the vesicle by the vesicular monoamine transporter (VMAT)
- Dopamine is converted to NE in the vesicle by dopamine- β -hydroxylase

- **Physiologic release occurs when**

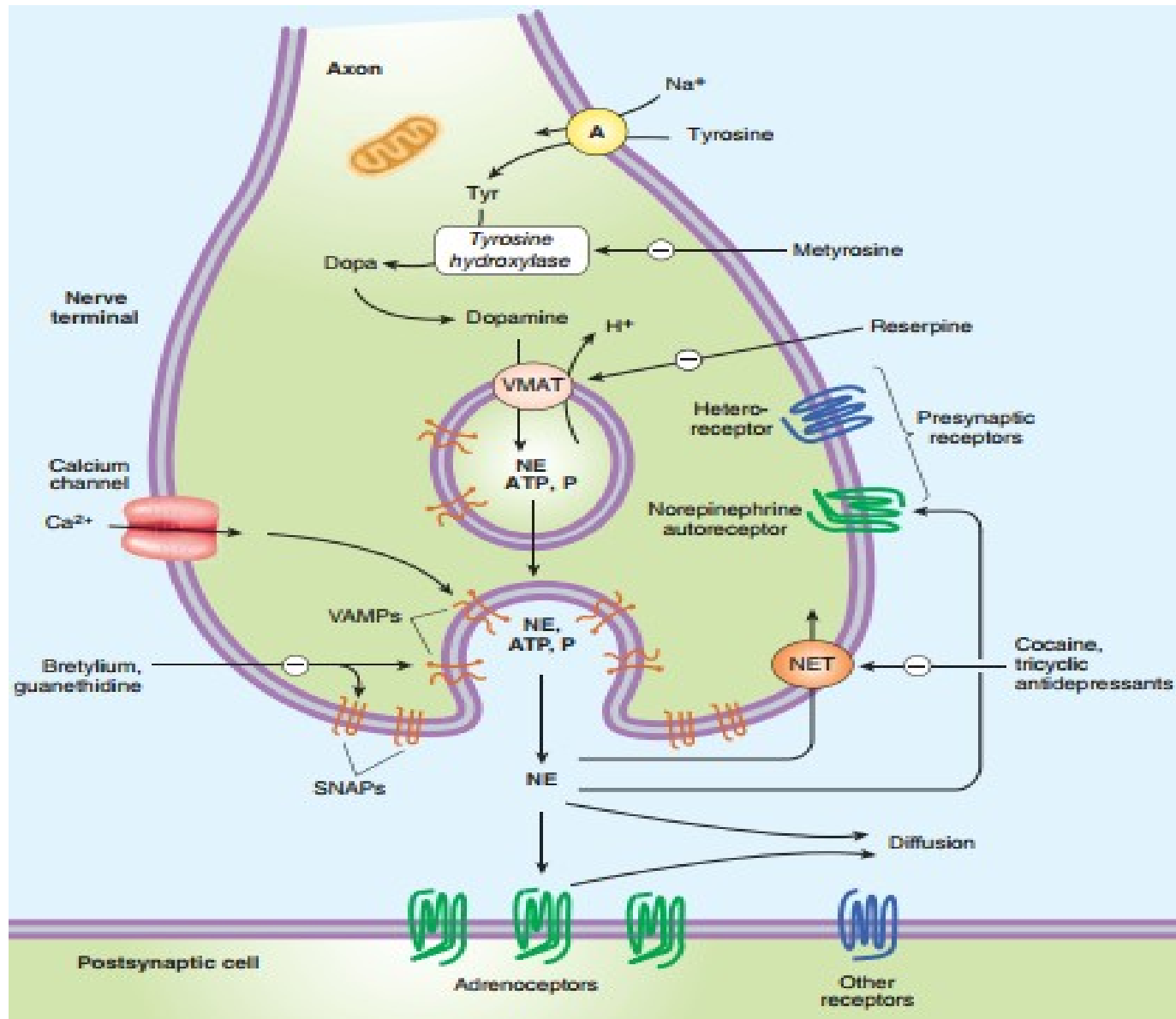
- AP opens voltage-sensitive calcium channels → increases intracellular calcium.



Removal of norepinephrine

1. Neuronal uptake (Reuptake 1)
 - primary mechanism for NE removal
 - transported into the cytoplasm of the terminal by the transporter followed by either
 - vesicular storage
 - enzymatic inactivation by mitochondrial MAO.
2. Diffusion from the synapse into the circulation
 - Ultimate enzymatic destruction in the liver & renal excretion.
 - The liver has two enzymes
 - catechol-O-methyl transferase (COMT) and Monoamine oxidase (MAO)
3. Active transport into effector cells (extraneuronal uptake)
 - uptake 2
 - followed by enzymatic inactivation by COMT

- In **CNS**
 - Mainly through metabolism by MAO.
- Two isoforms of MAO
 - MAO type A and
 - MAO type B.



Autonomic Receptors

- Cholinergic receptors
 - Muscarinic receptors (M_1, M_2, M_3, M_4, M_5)
 - Nicotinic receptors
- Adrenergic receptors
 - α -adrenoceptor - $\alpha_1(a_{1A}, a_{1B}, \text{ and } a_{1D})$
 - $\alpha_2(a_{2A}, a_{2B}, \text{ and } a_{2C})$
 - β -adrenoceptor - β_1, β_2 and β_3

Nicotinic receptors

- Ligand-gated ion channels
 - Can be stimulated by nicotine
 - Effects blocked by curare, plant poison, at NMJ
- **Subtypes**
 - N_N (neuronal nicotinic receptors)
 - N_M (muscular nicotinic receptors)
- **Location**
 - N_N -----CNS, adrenal medulla, autonomic ganglia
 - N_M ----- confined to the skeletal neuromuscular junction;

Muscarinic receptors

- G-protein coupled receptors
- M_1 , M_2 , M_3 are well characterized
- M_1 -receptors ('neural')
 - M_1 receptors are present in brain, exocrine glands, and autonomic ganglia
 - mediate **excitatory** effects by inhibiting K^+ conductance
- M_2 -receptors ('cardiac')
 - found in the heart, brain, autonomic ganglia, and smooth muscle.
 - exert **inhibitory** effects,
 - mainly by increasing K^+ conductance
 - by inhibiting calcium channels

- M₃-receptors ('glandular/smooth muscle')
 - smooth muscle, exocrine glands, brain, and endothelial cells
 - produce mainly **excitatory** effects →
 - stimulation of **glandular secretions** (salivary, bronchial, sweat, etc.)
 - **mediate relaxation** (mainly vascular) of smooth muscle
 - by release of nitric oxide from endothelial cells
- M4 receptors are present in **brain and autonomic ganglia**.
- M5 receptors are found in the **CNS**.

Adrenergic Receptors/ Adrenoceptors

- All belong to the superfamily of G-protein-coupled receptors.
- α_1 -adrenoceptors:
 - Vasoconstriction
 - relaxation of GI smooth muscle
 - salivary secretion
 - hepatic glycogenolysis

- α_2 -adrenoceptors:
 - inhibition of **transmitter release** (NA & Ach), platelet aggregation, **contraction of vascular smooth muscle**, inhibition of insulin release
- β_1 -adrenoceptors: increased cardiac rate & force, increase renin
- β_2 -adrenoceptors:
 - bronchodilatation, vasodilatation, relaxation of visceral smooth muscle, hepatic glycogenolysis and muscle tremor
- β_3 -adrenoceptors: lipolysis

- Two main classes of presynaptic receptors have been identified :
 - **Heteroreceptors**
 - are presynaptic receptors that respond to neurotransmitters, neuromodulators, or neurohormones released from adjacent neurons or cells.
 - For example, NE can influence the release of ACh from parasympathetic neurons by acting on $\alpha 2A$, $\alpha 2B$, and $\alpha 2C$ receptors, whereas ACh can influence the release of NE from sympathetic neurons by acting on M2 and M4 receptors.
 - **Autoreceptors**
 - are receptors located on or close to axon terminals of a neuron through which the neuron's own transmitter can modify transmitter synthesis and release.
 - For example, NE released from sympathetic neurons may interact with $\alpha 2A$ and $\alpha 2C$ receptors to inhibit neurally released NE.
 - Similarly, ACh released from parasympathetic neurons may interact with M2 and M4 receptors to inhibit neutrally released ACh.

Innervation of Various Organs By The SNS and PSNS

- Many organs are innervated by **both**
- Most of the time, SNS and PSNS act in **opposite direction but not always**
- In some tissues and organs, SNS and PSNS **complement each other**

CVS

- **Blood Vessels**

- Most vascular smooth muscle is innervated solely by SNS → VC

SNS

- **Blood vessels** to skin & mucous membranes:
Constriction
- **Blood vessels** to skeletal muscles: Dilatation
- Some blood vessels in the face, tongue, and urogenital tract are innervated **by both**

The PSNS → VD

The Heart

- Innervated by **both** but d/t **distribution**.
- **SNS** - innervates the sinoatrial (S-A) node, atria and ventricles.
 - Increase in rate (**+** chronotropic effect), in force of contraction (**+** inotropic effect), and in A-V conduction tissue (**+** dromotropic effect)
- **PSNS** - innervates in the S-A node, atria, and A-V conduction tissue.
 - do not innervate the ventricular muscle
 - **negative chronotropic and dromotropic effect**
 - decrease in the contractile force of the atria not on ventricular

The Eye

- The radial iris muscle (dilator pupillae),
 - innervated by SNS → **mydriasis**
- The circular iris muscle (constrictor pupillae)
 - innervated by PSNS → **miosis**
- Smooth muscle of ciliary body
 - innervated by PSNS → contraction of the ciliary smooth muscle body → accommodation for near vision.
- PSNS is dominant in the eye

Bronchial Smooth Muscle

- Innervated by **both divisions of the ANS**
- PSNS → **BC**
- SNS → **BD**

Gastrointestinal Tract

- Innervation is complex.
- SNS + PSNS + enteric nervous system (ENS)
- SNS **inhibit** gut motility & gland secretion & contracts sphincters.
- PSNS **stimulate** smooth muscle contraction & gland secretion

PSNS

- **Bladder:** contraction of detrusor muscles & relaxation of sphincter – facilitate urination

SNS

- **Bladder:** Relaxation of detrusor muscles & contraction of sphincter – Inhibit urination
- **Uterus:** Relaxation

Salivary Glands

- Both SNS and PSNS lead to an increase in the flow of saliva.
- However, different nature of the saliva
 - **SNS** → a sparse, thick secretion
 - **PSNS** → a profuse, watery secretion

The Adrenal Medulla

- **Only by SNS**
- The cells of the adrenal medulla, called chromaffin cells
 - homologous with SNS postganglionic
- Secretes two hormones: epinephrine & norepinephrine.
- Activation of the sympathetic system
 - is during stress, fear, or anxiety
 - accompanied by increased secretion, primarily of Epinephrine

Drugs Acting on the Autonomic Nervous System

- ❖ Cholinergic Drugs (Parasympathomimetics)
- ❖ Cholinergic Antagonists (Parasympatholytics)
- ❖ Adrenergic Drugs (Sympathomimetics)
- ❖ Adrenergic Antagonists (Sympatholytics)

Cholinergic Agonists (Parasympathomimetics)

- Drugs that **stimulate the parasympathetic nervous system (PSNS)**
- **Drugs having opposite function to SNS**
- Mimic the effects of the PNS
- Based on MOA two classes of cholinergic agonists
 - Direct acting (Receptor agonists)
 - Indirect acting (Choline esterase inhibitors)

Cholinergic Agonists –

1. **Direct-acting:** bind to and activate muscarinic or nicotinic receptors and include the following subgroups:

- A) **Choline Esters:**

- Natural :Ach
- Synthetic :Methacholine, carbachol,
- are agonists of either muscarinic or nicotinic receptors
- Betanechol: more selective to muscarinic

- B) **Alkaloids:** Pilocarpine, muscarine, arecoline, nicotine

□ are pure muscarinic agonists

2. **Indirect-acting:** Inhibit the action of **acetylcholinesterase enzyme**

- A. **Réversible:** Bind to cholinesterase for a period of minutes to hours

- E.g. neostigmine, physostigmine, edrophonium

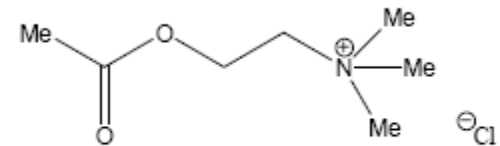
- B. **Irreversible:** bind to cholinesterase and form a permanent covalent bond

- E.g. Organophosphate compounds: Malathion, Parathion

» **The body must make new cholinesterase to reverse the effect**

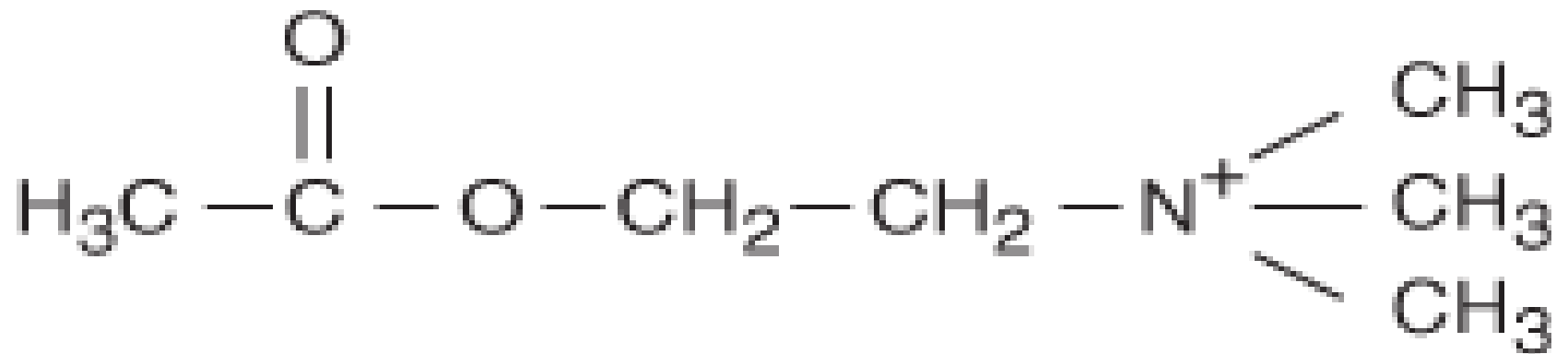
Acetylcholine

- ❖ Prototypical muscarinic and nicotinic agonist.
- ❖ Quick onset and short duration of action
- ❖ Has no/little clinical use b/c of
 - Its extreme short duration
 - Non specific/selective action
 - Rapid enzyme degradation
- ❖ Cannot be given orally (rapid hydrolysis → Low bioavailability)

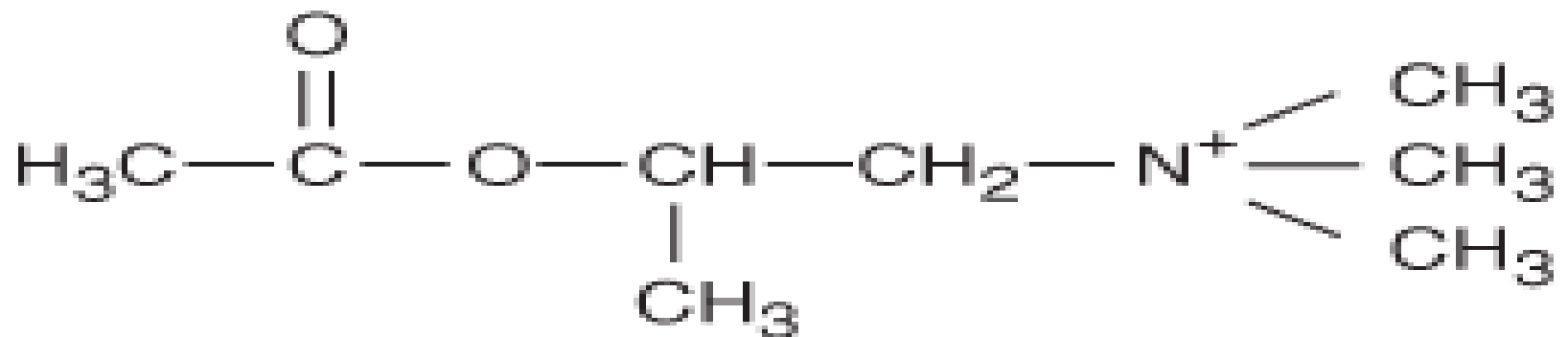


Acetylcholine chloride

- **Methacholine** differs from Ach
 - the addition of a methyl group at the β -carbon of ACh.
 - increases its **selectivity** for muscarinic receptors
 - renders methacholine **resistant** to the pseudo-ChE
 - decreases its susceptibility to AChE,
 - thereby increasing its potency



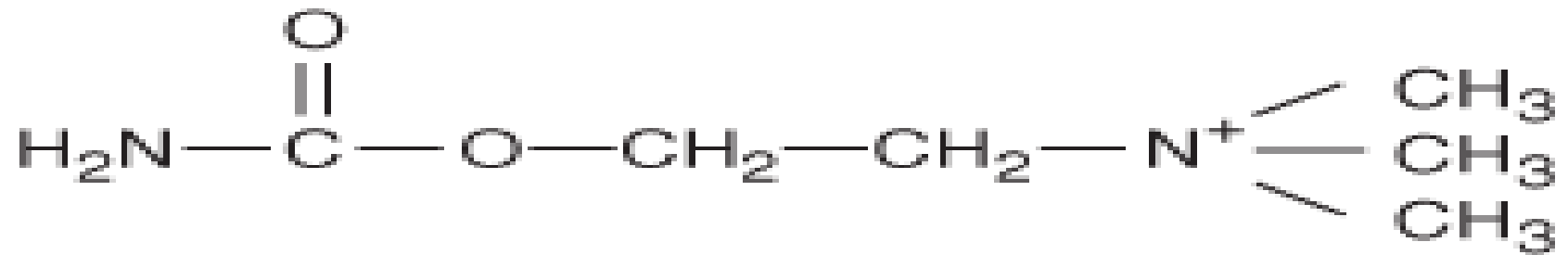
Acetylcholine



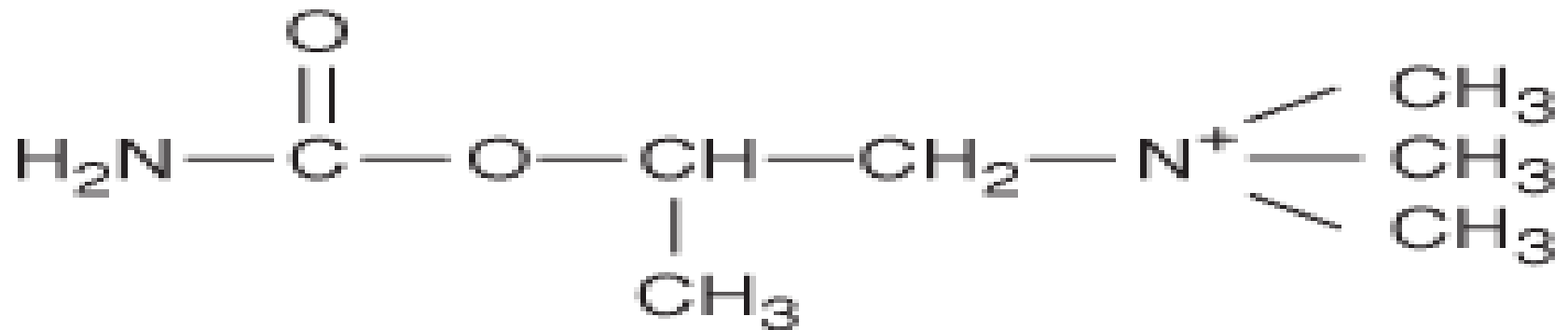
**Methacholine
(acetyl-β-methylcholine)**

Carbachol

- Differs from ACh
 - only in the substitution of a carbamoyl group for the terminal methyl group
 - completely resistant to degradation by cholinesterases
 - Does not improve its selectivity
- **Bethanechol** combines
 - Methacholine and carbachol
 - selective agonist of muscarinic receptors
 - resistant to degradation by cholinesterases.



**Carbachol
(carbamoylcholine)**



**Bethanechol
(carbamoyl-β-methylcholine)**

Pharmacokinetics

- **Choline esters**

- are hydrophilic as they retain the quaternary ammonium group
- are poorly absorbed
- poorly distributed into the CNS

- **Acetylcholine**

- very rapidly hydrolyzed ➔ limit systemic effects
- large amounts must be infused iv to produce response

Pilocarpine (M only)

- Exhibits muscarinic activity and is used primarily in ophthalmology.
- rapidly absorbed by the cornea of eye
- cross the blood-brain barrier
- **Actions:**
 - **Eye** (locally): Miosis and contraction of the ciliary muscle
 - **Secretions** (sweat, tears, saliva): increase

Organ System Effects of direct acting cholinomimetics

Organ	Response
Eye	
Sphincter muscle of iris	Contraction (miosis)
Ciliary muscle	Contraction for near vision
Heart	
Sinoatrial node	Decrease in rate (negative chronotropy)
Atria	Decrease in contractile strength (negative inotropy). Decrease in refractory period
Atrioventricular node	Decrease in conduction velocity (negative dromotropy). Increase in refractory period
Ventricles	Small decrease in contractile strength
Blood vessels	
Arteries, veins	Dilation (via EDRF). Constriction (high-dose direct effect)
Lung	
Bronchial muscle	Contraction (bronchoconstriction)
Bronchial glands	Stimulation
Gastrointestinal tract	
Motility	Increase
Sphincters	Relaxation
Secretion	Stimulation
Urinary bladder	
Detrusor	Contraction
Trigone and sphincter	Relaxation
Glands	
Sweat, salivary, lacrimal, nasopharyngeal	Secretion

Clinical uses of Cholinergic agonists

❖ **Major Challenge** in the clinical use of cholinergic agonists:

- ❑ Lack of selectivity for specific *subtypes* of muscarinic receptors.
 - This lack of specificity results in broad-ranging effect on different organ systems

❖ Clinical Uses include:

- ❑ Glaucoma (Pilocarpine is the first choice)
- ❑ GI Atony (Bethanechol most widely used for this purpose)
- ❑ Urinary retention
- ❑ Surgery of the Eye
- ❑ Alzheimers Diseases (**Oxotremorine widely used, since it cross BBB**)

Ophthalmologic Uses

- For the treatment of Glaucoma
 - Glaucoma - those conditions in which the intraocular pressure (IOP) is too high for the normal function of the optic nerve
 - Three major Categories
 - Primary Glaucoma
 - Secondary Glaucoma: e.g. Secondary to Inflammation, intraocular hemorrhage, trauma (e.g. post surgical glaucoma)
 - Congenital Glaucoma: - associated with hereditary diseases

Primary Glaucoma

- **Primary open-angle-glaucoma: -**
 - associated with **relative obstruction of aqueous humor outflow through trabecular meshwork.**
 - Also called wide angle glaucoma, chronic simple, glaucoma simplex
- **Primary angle-closure-glaucoma: -**
 - a condition in which **closure of the angle by peripheral iris results in a reduction in aqueous humor outflow**
 - Also known as narrow angle glaucoma, iris-block glaucoma, acute congestive glaucoma

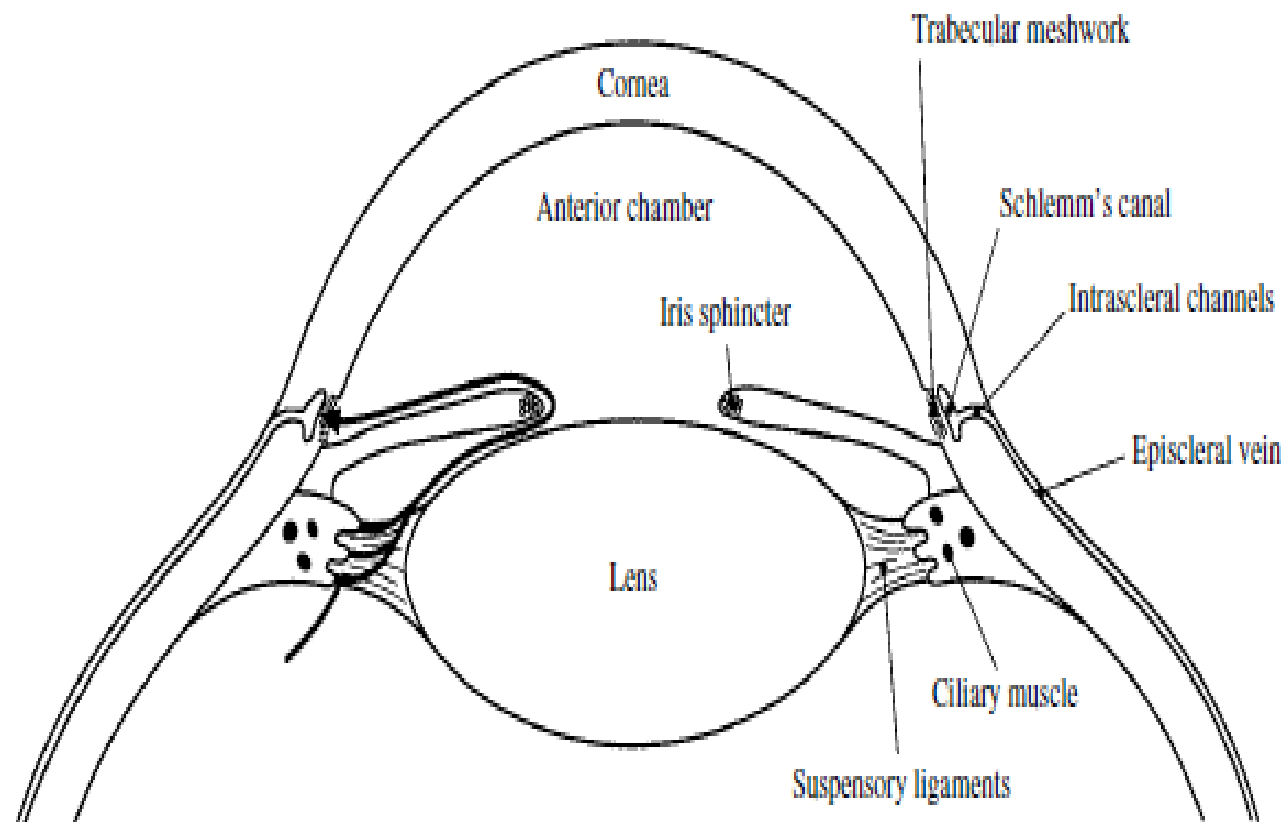


FIGURE 12.3

Diagram of eye depicting major pathway for outflow of aqueous humor (arrow) and ocular smooth muscles, which contract in response to parasympathomimetics or cholinesterase inhibitors (i.e., iris sphincter and ciliary muscle).

- Primary open-angle glaucoma
 - effectively treated with cholinomimetics
 - such as pilocarpine and carbachol
 - MOA:
 - **contraction** of the ciliary muscle
 - **Contraction** of the iris sphincter
 - **Pilocarpine** is the first choice given topically
 - **Carbachol** for resistant to pilocarpine
- **Angle-closure glaucoma**
 - **Cholinomimetics** may be used **to reduce IOP prior to surgery** (iridectomy)
 - Major treatment is surgical → iridectomy
- Cholinomimetics have been largely replaced
 - By topical β -blockers and prostaglandin.

- **Surgery of the Eye**

- Acetylcholine and Carbachol may be used as a miotic in surgery
- Acetylcholine for brief duration
- carbachol for a longer duration.

- To break adhesions b/n the iris and the lens

- Sequential administration of pilocarpine (miotic) and atropine (mydriatic)
- used to break iris-lens adhesions.

Gastrointestinal & Genitourinary

- **Post-operative; post-partum** urinary retention
 - Bethanechol is commonly used
- Paralytic ileus /postoperative ileus/
 - Is a condition characterized by atony or paralysis of the stomach or bowel following surgical manipulation
 - **Bethanechol** is used as smooth muscle stimulant
- Postoperative abdominal distention

- **Xerostomia**

- Pilocarpine is used following head and neck radiation treatments or Sjögren's (show-grins) syndrome
- Bethanechol is used as alternative → produces less diaphoresis

- **Cevimeline**

- a newer agonist with activity at M_1 , M_3
- used for the treatment of symptoms of **dry mouth with Sjögren's Syndrome.**
- Has a long-lasting **sialagogic** action and also enhances **lacrimal** secretions

- **Diagnosis of Bronchial Hyperreactivity**
 - Methacholine for testing bronchial hyperreactivity and asthma
 - an exaggerated airway contraction in patients who may be developing asthma

Adverse Effects

- Nausea, abdominal cramps, diarrhea, salivation, hypotension with reflex tachycardia, flushing, sweating, and bronchoconstriction, visual disturbance
- **Pilocarpine** can cross the blood-brain barrier and affect cognitive function.
- Potentially severe adverse effects can result from systemic administration
 - Generally **none should be administered** by intramuscular or intravenous injection.

Contraindications to Cholinergic agonists

- ❑ **Bronchial asthma**:-because they may induce bronchoconstriction and increase bronchial secretions
- ❑ **Peptic ulcer disease**:- b/c increase gastric acid secretion
- ❑ **Coronary insufficiency**:- because the hypotension produced will further compromise coronary blood flow
- ❑ **Mechanical** intestinal and urinary outlet obstruction

Indirect acting cholinomimetics/ Anticholinesterases

- Three chemical groups
 - ***Quarternary Alcohols***: - edrophonium, ambenonium
 - ***Carbamates*** : - Physostigmine, neostigmine, Pyridostigmine, Rivastigmine,
 - ***Organophosphates*** : - Echothiophate, Isoflurophate, Parathion, Malathion, tabun, sarin, and soman

Mechanism of Action of anticholinesterases

- **Inhibition of acetylcholinesterase**

- A simple competitive inhibition: -

- positively charged quaternary ammonium compounds bind to the anionic site and prevent ACh from binding.

- Acting either as a false substrate or directly attacking the esteratic site

- covalently modify the esteratic site and
- noncompetitively prevent further hydrolysis

Types of anticholinesterases

1. Reversible cholinesterase inhibitors

- Compete for Ach at the enzyme active site
- Form transient complexes with the enzyme

- **Quaternary alcohols**

- Act as simple competitive inhibitors,
- Binding and blocking access to active site
- Their positive charge allows them to bind to the anionic site at the reactive center
- without covalent modification of the site

Carbamates

- block access and tie up the active site serine
- Generally contain a tertiary or quaternary amine group
- Bind non-covalently to the anionic site of the enzyme
- Hydrolyzed by AChE like ACh
 - hydrolysis much more slowly
- Neostigmine and pyridostigmine
 - direct agonist activity at nicotinic



2. Irreversible cholinesterase inhibitors

- The organophosphates are irreversible cholinesterase inhibitors
- Bind to and undergo initial hydrolysis by AChE → phosphoryl moiety
 - Cleaved extremely slowly (requiring up to hundreds of hours) → 'irreversible'.
 - Full recovery require *de novo* biosynthesis of AChE

- Can also be reactivated by a strong nucleophile
 - an oxime, such as pralidoxime (2-PAM)
 - at least at an early stage of inhibition
- At later stages of inhibition,
 - the phosphoryl group on the enzyme will undergo *aging*
 - wherein an alkoxyl group is lost from the phosphoryl moiety
 - ➔ resistant to oxime attack
 - E.g. for isofluorophate

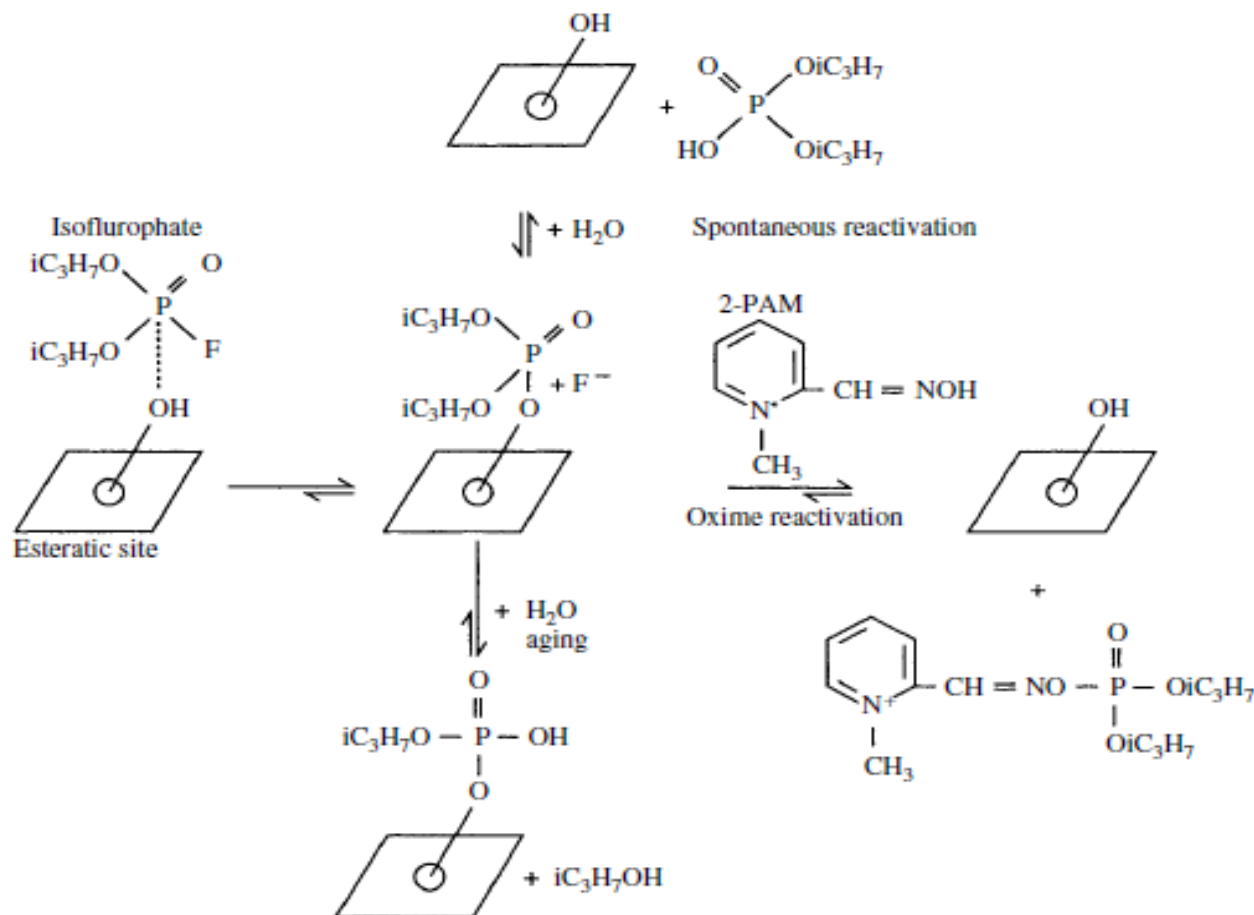


FIGURE 12.4

Isofluorophosphate reaction at AChE esteratic site, aging, spontaneous reactivation, and oxime reactivation. *Left*, the nucleophilic attack on the phosphorus of isofluorophosphate by the serine oxygen. This results in a stable phosphorylated enzyme intermediate, which undergoes dephosphorylation at a negligible rate (*top*). A more favorable reaction is the loss of an isopropoxy group, a process termed aging (*bottom*). This renders the phosphorylated enzyme resistant to dephosphorylation by an oxime. The original phosphorylated intermediate (*center*) will react with the nucleophilic oxygen of pralidoxime (2-PAM), resulting in dephosphorylation of the enzyme and formation of an oxime phosphonate (*lower right*).

Pharmacokinetics

- The tertiary amines
 - physostigmine, rivastigmine, tacrine, donepezil, and galanthamine
 - are well absorbed orally
- **Quaternary ammonium**
 - poorly absorbed after oral administration
 - Neostigmine and pyridostigmine orally active at larger doses are employed.

- Most of the organophosphates
 - are absorbed by all routes
 - high lipid solubility and low molecular wt
- Echothiophate is
 - quaternary ammonium organophosphorus
 - does not readily penetrate the skin

ORGAN SYSTEM EFFECTS

- Similar but not always identical to direct acting agents
- The lipid-soluble AchEIs,
 - cause generalized convulsions → followed by coma and respiratory arrest
- Eye, respiratory tract, GIT, urinary tract
 - Similar to direct-acting
- CVS
 - Negative chronotropic, dromotropic, and inotropic effects

- The fall in cardiac output
 - Bradycardia
 - Decreased atrial contractility
 - Some reduction in ventricular contractility
- Have less marked effects on BVs

Neuromuscular junction

- Increases strength of contraction
 - especially in myasthenia gravis and after neuromuscular blocking agents
- At higher concentrations → depolarizing neuromuscular blockade
 - persistent membrane depolarization
 - inactivation of voltage-gated sodium channels
 - Desensitizing

Clinical Uses

1. Treatment & diagnosis of Myasthenia Gravis

- **Myasthenia gravis**

- is an autoimmune dx in which antibodies recognize & destroy nicotinic cholinoreceptors on skeletal muscle
- Decrease the number of functional receptors → decreases the sensitivity of the muscle to Ach → Muscle weakness and rapid fatigue

- **Anticholinesterase**

- Increase the amount of Ach
- Neostigmine , Pyridostigmine, Ambenonium

2. Gastrointestinal and Urinary Bladder

- Paralytic ileus or urinary bladder atony.
- Neostigmine = drug of choice for treatment

3. Glaucoma

- Primary glaucoma
- Physostigmine, Demecarium, Echothiophate, Isoflurophate
- Topical application
 - presents the risk of systemic effects
 - can cause cataracts
- Largely replaced by topical β -blockers and prostaglandin derivatives
- Angle-closure glaucoma
 - Pilocarpine should be used rather than AChE inhibitors

4. Alzheimer's Disease

- A slowly developing neurodegenerative disease
 - produces a progress loss of memory and cognitive function.
 - result from the loss of cholinergic transmission.
- Tacrine, Donepezil, galantamine, and rivastigmine
 - Mild to moderate .
- Donepezil, rivastigmine, and galantamine
 - as effective as tacrine
 - do not share tacrine's hepatotoxicity

5. Reversal of Neuromuscular Blockade

- Reverse the neuromuscular blockade caused by non depolarizing blockers
- Neostigmine, pyridostigmine, edrophonium

6. Antimuscarinic Toxicity

- Physostigmine as
 - Reverses the central as well as the peripheral signs of muscarinic blockade
- However, physostigmine can produce cardiac arrhythmias
 - Should be considered as an antidote only in life-threatening cases

- **Adverse effects** of PA
- Poisoning
 - Accidental → agricultural and household insecticides
 - Chemical warfare → military nerve gases
 - Miosis, salivation, sweating, bronchial constriction, vomiting, and diarrhea
- Excessive inhibition can ultimately lead to a cholinergic crisis that includes:
 - Gastrointestinal distress (nausea, vomiting, diarrhea, excessive salivation),
 - Respiratory distress (bronchospasm & increased bronchial secretions)
 - Cardiovascular distress (bradycardia or tachycardia, A-V block, hypotension),

- Visual disturbance (miosis, blurred vision),
- loss of skeletal motor function (progressing through incoordination, muscle cramps, weakness, fasciculation, and paralysis).
- sweating, and
- CNS symptoms include agitation, dizziness, and mental confusion.
- Death
 - from paralysis of skeletal muscles required for respiration

Management of Carbamates and Organophosphate toxicity:

- ❑ Artificial respiration
- ❑ Decontamination
- ❑ Atropine (for muscarinic symptoms)
- ❑ Acetyl cholinesterase reactivate (Pralidoxime)
 - Pralidoxime is an antidote for organophosphate poisoning they should be used within a few hours following exposure, before aging has occurred, reactivation of cholinesterase is practically impossible, once being aged

Cholinoceptor-Blocking Drugs

i) Antimuscarinic drugs

- Parasympatholytic = block PSNS
- Drugs include: -
 - Naturally occurring alkaloids: Atropine (hyoscyamine), Scopolamine (hyoscine)
 - Synthetic and semisynthetic : - Ipratropium, Tropicamide, Benztropine, Pirenzepine, Telenzepine, Dicyclomine, Glycopyrrolate, Propantheline, oxybutynin, tolterodine

Mechanism of action

- Competitively antagonize the binding of ACh to muscarinic receptors.
- Have no intrinsic activity
- Produce effects **only by blocking the activation of muscarinic receptors**
- The magnitude of the response depends
 - on **the existing level of cholinergic activity** or on **the presence of muscarinic agonists**
- The nature of the response of an organ
 - depend on pattern of innervations.

Atropine

- The prototype, highly selective for muscarinic receptors
- does not distinguish b/n the M_1 , M_2 , & M_3
- **Other antimuscarinic drugs are moderately selective**
- The effectiveness of antimuscarinic drugs varies from tissue to tissue.
 - salivary, bronchial, and sweat glands **most sensitive tissues**
 - Gastric parietal cells is the **least sensitive**
- **Dicyclomine, oxybutynin, tolterodine**
 - nonselective smooth muscle relaxants

Organ system effects

- Tissues/systems affected depend on
 - The dose administered
 - The drug's PK
 - The sensitivity of muscarinic receptors

Central Nervous System

- Atropine vs Scopolamine
 - Scopolamine has prominent central effects → easily penetration of BBB
 - Atropine typical doses → have minimal effects
- Higher doses → excitation, agitation and hallucination

- **Scopolamine**
 - Sedation, amnesia, and drowsiness are common during the clinical use
 - Large doses of scopolamine → similar effects
- **Other tertiary amine compounds**
 - Similar effects

Eye

- Mydriasis
 - relaxation of circular muscle (mydriasis)
 - relaxation or weakening of ciliary muscle
- Cycloplegia
 - results in loss of the ability to accommodate for near vision
- Reduce lacrimal secretion
 - dry or "sandy" eyes with large doses

Cardiovascular system

- Low doses
 - a slight paradoxical bradycardia
 - blockade of **presynaptic muscarinic** receptors
- Higher doses
 - tachycardia by directly blocking PSNS
- Minimal effects on blood vessels
- **Respiratory system**
 - Bronchodilation and reduce secretion
- Potent inhibitors of secretions throughout the Res. system

Gastrointestinal Tract

- Inhibition of salivation → dry mouth and difficulty in swallowing
- Also inhibit gastric acid secretion & GI motility
- Pirenzepine and a more potent analog, telenzepine
 - reduce gastric acid secretion with fewer adverse effects than atropine

Genitourinary tract

- Can cause urinary retention
 - block excitatory effect of ACh on detrusor muscle of the bladder
- **Sweat glands**
 - Atropine inhibits sweating
 - Body temperature is elevated **at large doses**
 - But **in infants and children** even ordinary doses may cause "atropine fever."

Clinical uses

1. CNS disorders

- Parkinsonism
 - Treat tremor of parkinsonism
- In pts treated with antipsychotic drugs
 - used for preventing dystonias or parkinsonian symptoms
- Benztropine, biperiden, procyclidine, and trihexyphenidyl hydrochloride

2. Motion sickness

- Scopolamine
 - useful when the motion is **very stressful and of short duration**
 - Blockade of cholinergic sites in the vestibular nuclei and reticular formation
- When the motion is less stressful and lasts longer??

3. Uses in Ophthalmology

- To produce mydriasis and cycloplegia
- Cycloplegia used for
 - an accurate determination of the refractive error of the eye in uncooperative patients
- Mydriasis helps
 - ophthalmoscopic examination of the retina
- Atropine, scopolamine, cyclopentolate and tropicamide

4. Uses in Respiratory Disorders

- Ipratropium = inhalational B. Dilator in asthma and COPD
 - Negligible absorption from the airways & GIT
 - Most of the drugs are eliminated in the feces → systemic antimuscarinic effects

5. Uses in Anesthesiology

- atropine or scopolamine, Glycopyrrolate (im)
- To block excessive salivary and respiratory secretions
 - Induced by certain inhalation anesthetics
- **Newer inhalational anesthetics are less irritating** to the airways
 - routine use has declined

6. Gastrointestinal Disorders

- Treatment of peptic ulcer
- Antispasmodics to decrease pain of spasm
- Antidiarrheals
 - in common traveler's diarrhea
 - as adjunctive therapy in the trt **of irritable bowel syndrome**
 - Often combined with **an opioid antidiarrheal drug**

7. Urinary Disorders

- Propantheline, oxybutynin, dicyclomine
- Used for
 - nocturnal enuresis
 - urinary incontinence
- Oxybutynin
 - selective for M_3 receptors
 - is commonly used

8. Cholinergic Poisoning

- military nerve gases or mushroom poisoning
- Antimuscarinics /atropine/

9. Cardiovascular Disorders

- Atropine - initial treatment of patients with **acute myocardial infarction**
 - When excessive vagal tone causes sinus or nodal bradycardia

Adverse Effects and Poisoning

- Therapeutic effects vs undesirable effects
- Mydriasis and cycloplegia are
 - therapeutic effects in ophthalmology
 - adverse effects in reduce GI secretion/ motility
- General side effects at standard doses
 - xerostromia, blurred vision, photophobia, constipation, urinary retention, tachycardia
- CNS effects more apparent **after large doses**
 - headache, restlessness, ataxia, and hallucinations
- They also result in hot & flushed skin

CI and cautions

- CI in angle-closure glaucoma.
- Should be used with caution in untreated open-angle glaucoma
- Can aggravate reflux esophagitis
 - decreasing the tone of lower esophageal sphincter
- Infants and children
 - are especially sensitive to the hyperthermic action
- Should not be given in GI infections
 - slow gastric motility → retain MO

Ganglionic blocking drugs

- Drugs include:
 - Hexamethonium, Mecamylamine, Trimethaphan
- **Mechanism of action**
 - Block nicotinic receptors of both PSNS and SNS autonomic ganglia.
 - **non-depolarizing competitive antagonists**
- However, hexamethonium acts
 - by occupying sites in or on the nicotinic ion channel and not the receptor
- **Nicotine, Tetraethylammonium, and related drugs (eg lobeline)**
 - initially stimulates the ganglia
 - then blocks them

➔ depolarizing blockade

Pharmacokinetics

- **Mecamylamine**

- a secondary amine and the rest contain quaternary ammonium
- better absorbed from po
- penetrates BBB
 - produce CNS effects tremors, mental confusion, seizures, mania, and depression
- Its distribution is not confined to the extracellular space as that of quaternary cpds

Pharmacological Actions

- Produce non selective effects
 - block neurotransmission at ganglia
- The response obtained depends on
 - whether SNS or PSNS is dominant in controlling the function of the organ

Site	Effect of Ganglionic Blockade
<i>Tissues predominantly under parasympathetic (cholinergic) tone</i>	
Myocardium	
Atrium; S-A node	Tachycardia
Eye	
Iris	Mydriasis
Ciliary muscle	Cycloplegia
GI tract	Decrease in tone and motility; constipation
Urinary bladder	Urinary retention
Salivary gland	Dry mouth
<i>Tissues predominantly under sympathetic (adrenergic) tone</i>	
Myocardium	
Ventricles	Decrease in contractile force
Blood vessels	
Arterioles	Vasodilation; increase in peripheral blood flow; hypotension
Veins	Vasodilation; pooling of blood; decrease in venous return; decrease in cardiac output
Sweat glands ^a	Decrease in secretion

Clinical uses

- Limited clinical use
 - due to lack of selectivity

1. Hypertension

- Were once widely used in the management of **essential hypertension**
- Decline in their use
 - development of tolerance to these drugs
 - numerous undesirable side effects

con't

2. Treatment of **hypertensive crisis**

- Retain some usefulness
- Trimethaphan

3. Controlled Hypotension

- used to achieve **controlled hypotension** in plastic, neurological & ophthalmological surgery
- Commonly used in surgical procedures involving **extensive skin dissection**

Adverse Effects

- All pharmacological actions mentioned above occur as adverse effects.
- These effects are often intolerable except for acute use

- 2. A patient you are treating in the hospital has a hypertensive emergency, with blood pressure of 210/140 mm Hg. Of the following drugs, which would be most effective intravenously?

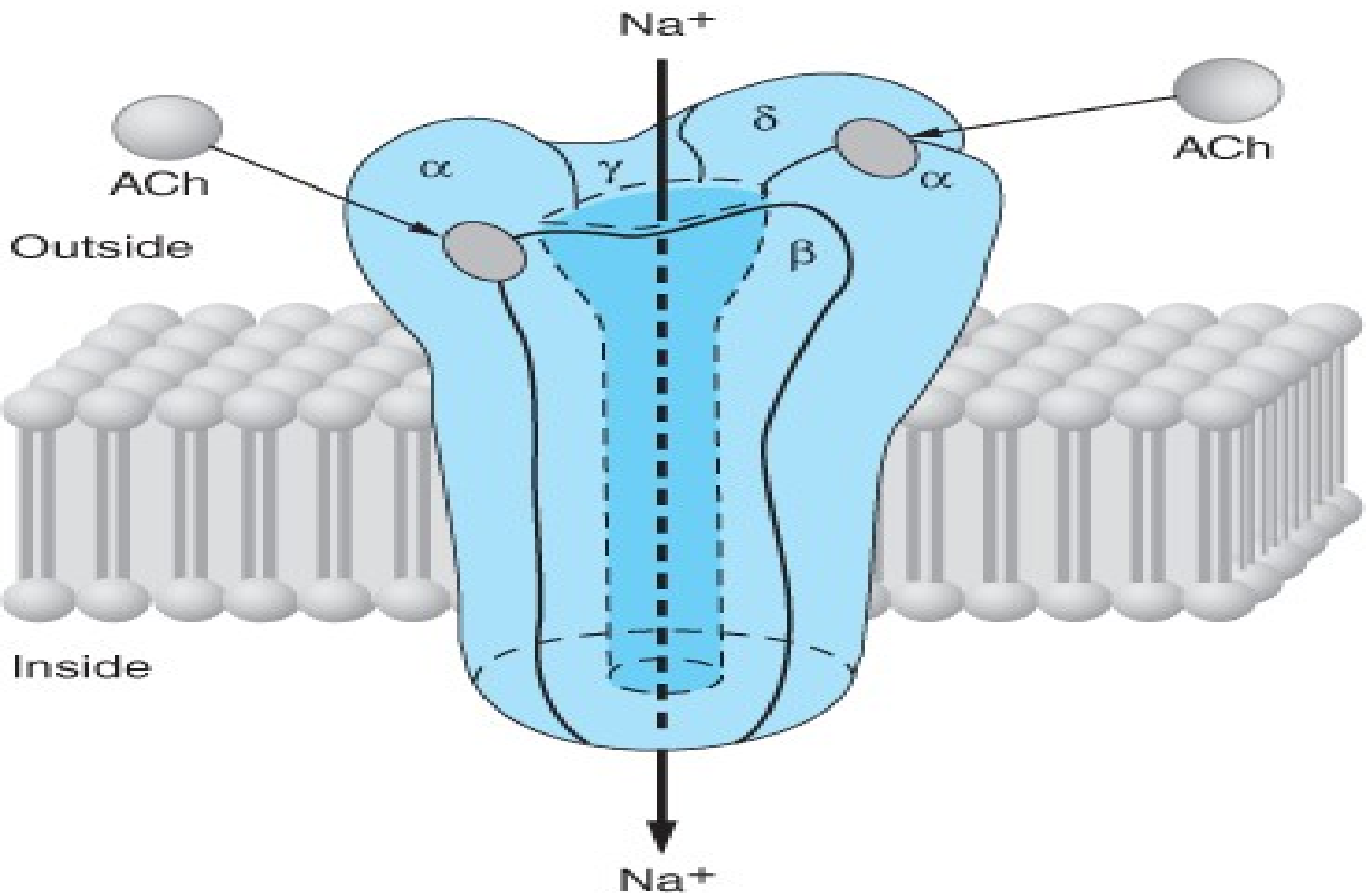
(A) Hydralazine
(B) Hydrochlorothiazide
(C) Trimethaphan
(D) Methyldopa
(E) Spironolactone

3. Ganglionic blocking agents are rarely used because of the numerous side effects they may produce. One such side effect is

(A) Increased stimulation of the genital tract
(C) Vasoconstriction
(D) Increased cardiac output
(E) Mydriasis

Neuromuscular Blocking drugs

- Interfere with Neurotransmission at neuromuscular end plate
- There are several types of neuromuscular receptors:-
 - Skeletal muscle
- They differ in subunit composition
- They differ in affinity for drugs
- Ligands at the receptor need only be cations
 - Almost all agonists and antagonists are quaternary ammonium compounds



Mechanism of action of Neuromuscular blocking Drugs (NMBDs)

- They are structurally related to ACh,
 - Contains a quaternary nitrogen group ($\text{N}^+(\text{CH}_3)_3$)
- As with ACh,
 - **the positive nitrogen atoms of NMBDs are attracted to the α subunits**
- Many NMBDs (**succinylcholine**, pancuronium and atracurium)
 - contain two quaternary ammonium cations
 - more potent than **monoquaternary amines**
- Monoquaternary amines
 - have only one permanent quaternary cation and one tertiary amine
 - rocuronium, **tubocurarine** & vecuronium

Types of NMBDs

- Depolarizing vs non-depolarizing drugs

1. Non-depolarizing NMBDs

- Drugs include: -
 - Tubocurarine, Pancuronium, Metocurine, Atracurium, Vecuronium, Mivacurium, Rocuronium, Doxacurium

MOA

- Antagonize the action of Ach
 - in a competitive manner
 - Bind to one or both α -subunits → prevents access to Ach
→ gradual reduction in AP
 - Neuromuscular block only becomes evident when **70-80% of receptors are occupied**
 - To produce **complete block**, at least **92% of receptors** must be occupied
 - do not produce conformational changes in the receptor

2. Depolarizing NMBDs

- Are agonists at ACh receptors
- Drug include: Succinylcholine (clinical use) and Decamethonium
- Succinylcholine
 - is effectively two Ach molecules joined through the acetate methyl groups
 - a dysfunctional agonist
 - The two quaternary ammonium radicals bind to the two α subunits of one nicotinic receptor, and depolarization occurs.

- Activation of the ACh receptors → membrane depolarization → opening then closing → inactivation of Na channel
- The membrane potential must be reset
 - before the Na channels reactivated
- This is a very rapid process with ACh (1 ms) b/c of AChE
- Succinylcholine is not metabolized by AChE
 - a prolonged activation of the nAChR
 - inactivation of sodium channels → flaccid muscle
- Depolarization block
 - also called Phase I/accommodation block
 - is often preceded by muscle fasciculation

Pharmacological Actions

- **Skeletal Muscle Paralysis**
 - Motor weakness progresses to a total flaccid paralysis
 - Small, rapidly moving muscles (the eyes, jaw, and larynx) relax → limbs and trunk → the intercostal muscles → diaphragms are paralyzed → respiration ceases
- Recovery of muscles usually occurs in the reverse order to that of their paralysis
- Percentage of receptors that may be blocked before paralysis is
 - greatest in the muscles of respiration
 - great in muscles of coarse movement (e.g. gastrocnemius)
 - least in the muscles of the face and eyes

Clinical uses

- Adjuvant in surgical anesthesia
 - To achieve **muscle relaxation and abolition of reflexes sufficient** for deep surgery
 - Facilitates operative manipulations.
 - If anesthetics are used alone → dangerous levels of anesthesia are required.
 - Also of value in various orthopedic procedures
 - the correction of dislocations & alignment of fractures
- **To facilitate intubations**
- NMJBD of short duration often are used to facilitate intubation for
 - laryngoscopy, bronchoscopy/esophagoscopy
 - in combination with a general anesthetics

ADRs

- Tubocurarine has several drawbacks:-
 - **It blocks ganglia** – loss of control of BP
 - **It affects muscarinic receptors** – loss of para-sympathetic control of **heart rate- tachycardia**
 - **It releases histamine** → hypotension, broncho-constriction and anaphylaxis.
 - **Its duration of action is over 30 min** – too long for short procedures such as intubation.
 - It is excreted unchanged in the **urine with some biliary excretion.**
 - Its duration of action is increased in renal failure and in elderly patients

Drugs	Duration 5% Recovery	Ganglion block?	Vagolytic?	Histamine release?
Tubocurarine	35 min	Yes	Yes	Yes
Pancuronium	25 min	No	Little	No
Metocurine	30 min	No	No	Yes
Atracurium	15.5 min	No	No	Yes
Vecuronium	12 min	No	Little	No
Mivacurium	8 min	No	No	Yes
Rocuronium	28 min	No	No	No
Doxacurium	1-3 hrs	No	No	No

Succinylcholine

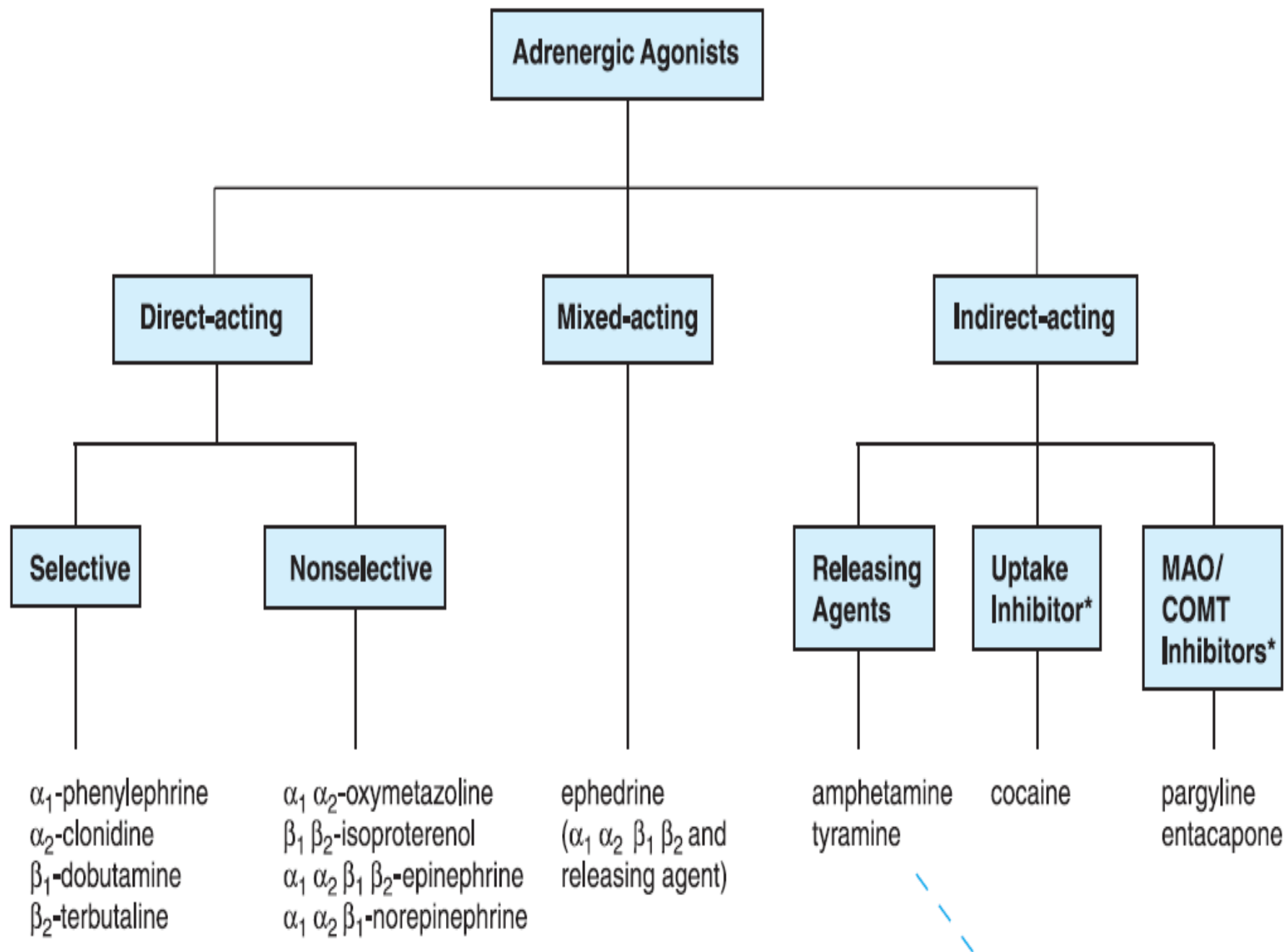
- Cardiovascular effects
 - Stimulation of mAChRs in SA node produces **bradycardia**
 - esp in pts with a high vagal tone (e.g. children).
 - In adult, bradycardia is seen more commonly after repeated increments
 - Ventricular arrhythmias may develop.
- Muscle pain
 - **Postoperative muscle pain**
 - More common in the young and healthy with a large muscle mass.

- **Prolonged paralysis**

- Patients with **congenital deficiency of plasma cholinesterase**

- **Hyperkalemia**

- Can **release K^+ rapidly from intracellular sites**
- Release of potassium may be exaggerated in muscular dystrophies



Adrenergic agonists

- ❑ Also known as adrenomimetic drug.
 - Mimic the effects of adrenergic sympathetic nerve stimulation
- ❑ Classified based on mechanism of action as:
 1. **Direct acting**
 - Act directly on one or more of the adrenergic receptors
 2. **Indirect acting**
 - Increase the availability of nor epinephrine.
 - By releasing or displacing NE from sympathetic nerve stores.

Eg, ***Amphetamine, Tyramine***

Adrenergic agonists --

- By blocking the transport of NE into sympathetic presympattic neurons.
 - » **Cocaine** , Tricyclic antidepressants (TCAs)
- By blocking the metabolizing enzymes, MAO (Pergyline) or COMT (Entacapone).

3. Mixed acting sympathomimetics

- Indirectly release norepinephrine and also directly activate receptors Eg. **Ephedrine**, **pseudoephedrin**, **Dopamine**.

Direct acting Adrenergic agonists --

❖ Classified based on chemistry as:

- Catecholamines
- non- catecholamines

A. Catecholamines:

- Naturally occurring:-Norepinephrine epinephrine and Dopamine
- Synthetic : - Isoproterenol (isopropyl norepinephrine)

□ They have the following characteristics:

- High potency
- Rapid enzymatic inactivation by **MAO & COMT** as well as **neuronal & non-neuronal uptake**
- Therefore, they have short duration when given parenterally and are inactive orally
- Poor ability to pass the CNS

B. Non-catecholamines

- ***Non-catecholamines*** are adrenergic agonists lacking the catechol hydroxyl groups
- They are not inactivated by COMT
 - Therefore, they are of longer duration,
- Can be given orally
- They include
 - phenylephrine
 - ephedrine
 - terbutaline, and amphetamine

Nonselective(alpha and beta) Adrenergic Agonists

- Act nonselectively both at alpha and beta receptors

- Epinephrine

- Norepinephrine

- Dopamine

A. Epinephrine/Adrenaline

- Potent stimulant of α and β receptors
- E agonist on α_1 , α_2 , β_1 , β_2 , β_3
- Actions on target organs

☐ Blood Pressure

- A small dose → a fall in mean and DBP with little or no effect on SBP
 - Due to the net decrease in total PR → predominance of VD in the skeletal muscle BVs

❑ Heart

- Direct myocardial stimulation- +ve inotropic
- Increased heart rate -+ve chronotropic
- Increased conduction- +ve dromotropic
- Cardiac output is markedly increased → largely responsible for increase in SBP

❑ Blood vessels

- NE & E act on post junctional α_1 - and α_2 → VC
- On pre-junctional and endothelial α_2 -→ VD
- Vasoconstriction in skin and mucus membranes – α effect
- The BVs in visceral organs contain predominantly α , some β_2 → VC

Epinephrine ---

- Vasodilation in skeletal muscles- (β -2 effect) ; Low doses → VD, Larger doses → VC
 - Counterbalanced
- Vasoconstriction which leads to increased peripheral resistance
- Insignificant Reflex Bradycardia

❑ Respiratory

- Powerful bronchodilator(β -2 mediated effect)
- Nasal decongestion (alpha1 mediated effect)
- Inhibition of antigen-mediated production of inflammatory mediators(β -2 mediated effect)

Epinephrine ---

❑ Smooth Muscles

- Radial (dilator) muscle of the iris
 - Contains α -adrenoceptors $\rightarrow$ produce mydriasis
- Relaxation of GIT (β_2 , α_2) and Urinary tract smooth muscle (β_2 , α_1)
- In the pregnant uterus there is inhibition of tone and contractions via β_2
 - β_2 -adrenoceptors agonists used to delay premature labor

❑ Metabolic effects

- Increases concentration of glucose (hyperglycemia)
 - Increase Hepatic & Skeletal Muscle Glycogenolysis (β_2)
 - Decreases uptake of glucose by peripheral tissue
- Increases free fatty acid concentration in blood (Lipolysis)

CNS

- E in therapeutic doses mildly stimulates the CNS
 - restlessness, apprehension, headache, and tremor

B. Norepinephrine

- Pharmacological effects differ with Epinephrine in ratio of their effectiveness in stimulating α & β_2 receptors
- Epinephrine and NE are equipotent at β_1
- NE is potent at α_1 , α_2 & has little action on β_2

Cardiovascular System

Blood vessels

- Skin and mucous membranes BVs - $\alpha \rightarrow VC$
- The BVs in visceral organs contain predominantly α , some $\beta_2 \rightarrow VC$,
- The BVs in skeletal muscle contain both α and $\beta_2 \rightarrow VC$

Effect on Blood Pressure and heart

- A net increase in the total PR $\rightarrow \uparrow$ in BP, Due to VC
- On heart complex b/c of
 - the dynamic interaction of the direct effects of NE and baroreceptor reflexes
 - NE –directly stimulatory (positive inotropic and chronotropic)
 - The reflex – inhibitory (opposite to the direct effect)

Eye

- Radial (dilator) muscle of the iris
 - Contains α -adrenoceptors → produce mydriasis

- **GI and bladder**

- The trigone and sphincter; contain α_1 -receptors
 - are contracted by NE

➔ Inhibit the voiding of urine

- GI muscle is relaxed
- Resting tone and peristaltic mov'ts are reduced due to
 - the inhibitory effect of β_2 , and
 - inhibition of release of ACh (via α_2)

- Contraction of the sphincters (via α_1)

Uterine muscle → NE contracts the uterus.

CNS

- NE also have minor CNS stimulant
 - do not easily cross the BBB

- **CVS**
- IP → act on β_2 → VD
- increases the HR and SV more than does epinephrine.
 - Partly due to its ability to **decrease mean BP** that reflexively diminishes vagal activity, and
 - Partly due to its direct action on the heart.
- **Bronchial smooth muscle:** IP → BD (via β_2) while NE has a relatively weak action
- IP relax (via β_2) the **uterus**.
- Detrusor muscle → **relaxed by IP**.
- IP has minor CNS stimulant: do not easily cross the BBB
- IP → relatively **weak hyperglycemia**.
- **IP is the most potent agonist** followed by E and NE

C. Dopamine

- ❖ Neurotransmitter in the brain.
- ❖ Precursor in the synthesis of NE
- ❖ Stimulates dopaminergic, beta-1 and alpha-receptors.
- ❖ Use as a drug
- ❖ Dopamine exerts its cardiovascular actions by
 1. Releasing NE from adrenergic neurons
 2. Interacting with α -and β -ARs
 3. Interacting with specific dopamine receptors

Dopamine----

❖ Effect is dose dependant:

- ❑ In **small doses 2-5µg/kg/minute**, it stimulates D1-receptors in renal, mesenteric and coronary vessels leading to **vasodilatation**
 - Dopamine Increases renal blood flow, GFR and causes natriuresis
 - Renal vasoconstriction occurs in CV shock due to sympathetic over activity

Dopamine

❑ Moderate dose (5-10 µg/kg/minute)

- stimulates $\beta 1$ -receptors in heart producing positive inotropic and chronotropic actions
- Releases Noradrenaline from nerves by $\beta 1$ -stimulation
- Does not change TPR
- Great Clinical benefit in CV shock and CHF
- septic shock(dopamine is preferable)

❑ High dose (10-30 µg/kg/minute)

- stimulates vascular adrenergic $\alpha 1$ -receptors
- vasoconstriction and decreased renal blood flow
- Can exacerbate CHF and CV shock

Pharmacokinetics

- Epinephrine and Norepinephrine
 - Are not effective after orally → rapidly conjugated and oxidized in the GI mucosa and liver
 - Inhalation is locally effective
 - Not usually given IV
 - Absorption from sc slow b/c of local VC
 - Absorption is more rapid after im
 - Rapidly inactivated in liver by COMT and MAO

- **Isoproterenol**

- Is readily absorbed given parenterally or as an aerosol
- Primarily metabolized in the liver and other tissues by COMT
- Relatively poor substrate for MAO
- Is not taken up by sympathetic neurons to the same extent as E & NE
- Longer duration of action

- **Dopamine**

- Substrate for both MAO and COMT → ineffective orally
- Dopamine HCl is used only iv
- brief duration of action like E,
 - The rate of administration can be used to control the intensity of effect.
- Administered initially at a rate of 2 to 5 mg/kg per minute; this rate may be increased gradually up to 20 to 50 mg/kg per minute

Epinephrine --

❑ Clinical uses

- Drug of choice for allergic reactions(Including Anaphylactic shock)
- Prolong action of local anesthetics
- Broncho spasm
- Cardiogenic arrest
- **Epistaxis**; Epinephrine/ α -agonists used to stop bleeding from nasal mucosa
- topical *hemostatic agent*- control of local hemorrhage
- open angle glaucoma- promotes an increase in the outflow of aqueous humor

❑ Adverse effects

- Cerebral hemorrhage
- Cardiac arrhythmias
- Rapid heart rate/Tachy cardia

Norepinephrine

- Currently very little therapeutic use
- Infused iv to combat systemic hypotension
 - during spinal anesthesia or other hypotensive conditions
- Not used to combat the hypotension
 - due to most types of shock (Cardiogenic shock, septicemic shock)
 - Reflex VC generally is intense and perfusion of organs may be jeopardized by NE
 - Vasodilating drugs improve blood flow and decrease cardiac work

Isoproterenol

- Used in emergencies to stimulate heart rate in bradycardia or heart block.
- Its use in asthma and shock has been discontinued
 - Dev't of more selective sympathomimetics.
- **ADR**: Palpitations, tachycardia, headache, and flushing are common.
- Cardiac ischemia and arrhythmias
 - particularly in patients with underlying coronary artery

Dopamine

- Cardiogenic shock b/c of MI or CHF
- Septic shock to improve renal circulation
- The other catecholamines have disadvantages:
 - IP a powerful chronotropic agent and can greatly increase myocardial oxygen demand
 - NE intensifies peripheral vasoconstriction
 - E increases heart rate and may predispose the heart to dangerous arrhythmias.

- An **advantage of using dopamine in shock** is that
 - Its **inotropic action increases CO** while dilating renal BVs and thereby increasing renal BF
- It causes less increase in heart rate than does isoproterenol

❑ **Adverse effect:**

- tachycardia, anginal pain, headache, hypertension, and peripheral vasoconstriction.

Non selective($\alpha 1$ and $\alpha 2$) agonists

- ❑ Drugs which directly activate both $\alpha 1$ and $\alpha 2$ Receptors.
- ❑ Drugs include
 - Xylometazoline
 - oxymetazoline
- ❑ Use- **Topical decongestants** because of their ability to **promote constriction of the nasal mucosa.**
- ❑ Adverse effect:
 - oxymetazoline may cause **hypotension** (large doses)-may be due to $\alpha 2$ agonist effect

Selective α_1 -Adrenergic Agonists

- Include: Phenylephrine, methoxamine, *Midodrine*, Mephentermine, metaraminol

Phenylephrine

- selective α_1 receptor agonist
- **Increase total peripheral resistance** (inc. SBP & DBP) through their vasoconstrictor action .
- **Not inactivated by COMT** and has a much **longer duration of action** than the catecholamines.
- **Use-** treatment of glaucoma
 - **Local nasal decongestant**
 - To **overcome hypotension** induced by some general anesthetic agent
 - raise the blood pressure (followed by reflex bradycardia)

α 2-Selective Agonists

- ❑ Activation of presynaptic α 2 receptors
 - Inhibits the release of norepinephrine from PNS
 - Inhibits sympathetic nervous system activity and leads to a fall in blood pressure in CNS .
- ❑ Blockade of α 2 receptors can increase sympathetic outflow
 - Potentiate the release of norepinephrine from nerve endings
 - Increase BP
- **Drugs include:** Clonidine, Brimonidine, Guanfacine, Guanabenz, Methyldopa, Apraclonidine

Clonidine

- ❖ Agonist to **presynaptic α_2 adrenoceptors** in brain, stimulation suppresses sympathetic outflow and reduces blood pressure
- ❖ Activates peripheral presynaptic autoreceptors on adrenergic nerve ending mediating **negative feedback suppression of noradrenaline release**
 - Clonidine reduces BP
- ❖ Overdose stimulates peripheral **postsynaptic α_1 adrenoceptors** & cause **hypertension by vasoconstriction**
- ❖ Use- treatment of systemic hypertension.
- ❖ **Adverse effects:**
 - Dry mouth
 - Sexual dysfunction
 - Bradycardia
 - Withdrawal syndrome (**rebound hypertension**)

Methyldopa

- ❖ Centrally acting antihypertensive agent.
- ❖ Metabolized to α -methylnorepinephrine in the brain
 - activate central α_2 receptors and lower blood pressure in a manner similar to that of clonidine.
- ❖ **Safe and preferable anti-hypertensive agent during pregnancy**

β_1 -Adrenergic Agonists

Dobutamine

- ❑ Is a synthetic dopamine analog and selective β_1 -agonist.
- ❑ On the heart, it produces a more **pronounced positive inotropic effect than its chronotropic effect** when compared to dopamine.
- ❑ It produces renal and mesenteric vasodilatation (D_1 -receptors) similar to dopamine
- ❑ **Therapeutic use** of dobutamine is based on its ability to increase cardiac output via the positive inotropy with little effect on heart rate and myocardial oxygen consumption
- ❑ Hence, it is used in **cardiogenic shock and decompensated heart failure**

β_2 adrenergic receptor agonists

- ❑ Are selective β_2 adrenergic receptor agonists with a little effect/no effect on β_1 cardiac receptors
- ❑ Hence, they have the advantage of producing bronchodilation without cardiac stimulation
- ❑ Two types:
 - Short acting :Terbutaline, albuterol (salbutamol),
 - Long acting :Salmeterol ,Formoterol ,ritodrine
- ❑ They are given orally, IV or by inhalational and possess no CNS stimulation
- ❑ Effects:
 - Uterine relaxation
 - Bronchodilation
- ❑ uses
 - bronchial asthma and bronchospasm associated with bronchitis a
 - Delay delivery in premature labor and in threatened abortion; ritodrine

β_2 adrenergic receptor agonists)

Albuterol (Salbutamol

- ❑ Inhalation- produce significant bronchodilation within 15 minutes
 - Duration for 3 to 4 hours.
- ❑ Oral administration-onset delayed by 1 to 2 hours.
 - Oral albuterol has the potential to delay preterm labor.
 - CNS and respiratory side effects are sometimes observed
- Has no advantage than inhalation only cost

Ritodrine

- β_2 receptor agonist developed specifically for use as a uterine relaxant.
- Use: To arrest premature labor (intravenously).

Salmeterol

- ❑ Highly lipophilic and has a sustained duration of action
- ❑ This long action reflects binding to a specific site within the B2 receptor that allows for its prolonged activation.
 - may also have antiinflammatory activity.
- ❑ Since onset of action of inhaled salmeterol/formeterol is relatively slow, it is not suitable monotherapy for acute breakthrough attacks of bronchospasm.
 - Not given based on PRN bases rather on regular bases for **chronic management of asthma**

Indirectly Acting Adrenergic Drugs

Tyramine

- ❑ By-product of tyrosine metabolism in the body
- ❑ Found in high concentrations in some foods such as **cheese** and **fish**.
- ❑ Readily metabolized by MAO in the liver.
 - MAO inhibitors in conjunction with tyramine-containing foods may lead to **rapid release of NE & severe hypertension**
- ❑ Has an indirect sympathomimetic action caused by the release of stored catecholamines.
 - It enters synaptic vesicle and causes **displacement & release of NE & normally degraded by MAO**

Amphetamine

Amphetamine

- causes release of NE from adrenergic nerve endings
- blocks its reuptake into the cytoplasm of the nerve terminal.
- has potent peripheral effects on α_1 , α_2 and β_1 receptors, but not β_2 receptors.
- potent CNS stimulant which is orally effective.

CNS effects

- Stimulating the CNS.
- CNS effects include :
 - Increased alertness;
 - Decreased sense of fatigue
 - Elevation of mood, with increased initiative,
 - Self-confidence, and
 - Ability to concentrate; elation and euphoria;
 - Performance of simple mental tasks is improved
- Finally, performance is reduced by fatigue and lack of sleep.

Mixed-acting Agent

Ephedrine

- ❑ Chemically related to EP and stimulates release of NE
- ❑ It is not a substrate for COMT or MAO & hence has long duration of action
- ❑ It activates β_2 as well as α - and β_1 -adrenergic receptors
- ❑ It is used to treat mild cases of asthma
- ❑ It crosses BBB giving rise to CNS stimulant action
- ❑ It is now replaced by more selective β_2 agonists

Drugs Acting on the Adrenergic Receptor Subtypes

	α - 1	α - 2	B- 1	B- 2
Agonists	Phenylephrine, Methoxamine	Clonidine, MethylDopa, Oxymmetazoline	Dobutamine, Isoproterenol, Terbutaline	Salbutamol, Terbutaline, Isoetharine
Antagonists	Prazosin, Terazosin, Doxazosin Phentolamine, Phenoxybenzamine	Yohimbine. Phentolamine, Phenoxybenzamine	Propranolol, Pindolol, Atenolol, Metoprolol, Timiolol	Propranolol, Pindolol, Butoxamine, Timiolol

- **Adrenaline** stimulates all the four receptor subtypes.
- **Noradrenaline** stimulates both alpha receptors and beta1 but has very poor affinity for beta2receptors.
- **Labetalol & Carvedilol** blocks all beta receptors as well as some alpha receptors.

Adrenergic Antagonists /Sympatholytics

- Classification according to MOA:
 1. Direct-acting
 - Adrenergic receptor antagonists
 - Alpha receptor blockers- limited use
 - Beta receptor blockers- widely used
 - Selective & non-selective
 2. Indirect-acting
 - Act by inhibiting:
 1. Synthesis of NE- metyrosine
 2. Storage of NE- reserpine
 3. Release of NE- bretylium, guanethidine
 - Used only to treat HTN

Direct-acting Adrenergic antagonists

□ **α -antagonists**

- Selective α_1 blockers
 - Prazosin,
 - Terazosin,
 - Doxazosin
- Non-selective α blockers
 - » Phenoxybenzamine
 - » Phentolamine

□ **α_2 antagonists**

- » Yohimbine,
- » Tolazoline,
- » Rauwolscine

□ **β -antagonists**

- Selective β_1 blockers
(Cardioselective)
 - » Acebutolol, alprenolol,
 - » Atenolol, betaxolol,
 - » Celiprolol, esmolol,
 - » Metoprolol, Nebivolol
- Non-selective β blockers
 - » Carteolol,
 - » Propranolol,
 - » Penbutolol,
 - » Pindolol,
 - » Timolol
- Mixed antagonists (blockers)
 - Block beta1, beta2, alpha1,
Carvedilol

α -antagonists

Prazosin

- ❖ **selective for α_1 receptors** and typically 1000-fold less potent at α_2 receptors.
 - leads to a fall in **peripheral vascular resistance** and in venous return to the heart.
- ❖ Unlike other vasodilating drugs, **administration of prazosin usually does not increase heart rate.**
- ❖ Due to its **shorter duration of action**, the twice-daily dosing requirement for prazosin is a disadvantage compared with newer α_1 receptor antagonists.
- ❖ It has bioavailability of about 50% to 70% after oral administration.

Clinical Uses of Alpha Blockers

❑ Hypertension

- Hypertensive Emergencies,
 - α - Adrenoceptors antagonists are most useful when there is **excess of circulating α -agonists**: epinephrin or norepinephrin

❑ Pheochromocytoma

- Pheochromocytoma is a tumor usually found in the adrenal medulla that releases a mixture of epinephrine and norepinephrine.
 - Patients have many signs of **catecholamine excess**, including hypertension, tachycardia and arrhythmias.

Clinical Uses of Alpha Blockers

Urinary Obstruction(BPH)

- Due to partial reversal of smooth muscle contraction in the enlarged prostate and in the bladder base.

Erectile Dysfunction

- A combination of **phentolamine** with the nonspecific smooth muscle relaxant papaverine, when injected directly into the penis, may cause erections in men with sexual dysfunction.

- ❑ **Orthostatic-hypotension and syncope:-** causing dizziness, lightheadedness, and even syncope (fainting)
 - most common side effect at first dose (first-dose effect)
 - First dose should be small (≤ 1 mg) and administered at bed time
 - Advise the patient to sit or lie down
- ❑ **Reflex tachycardia** - by triggering the baroreceptor reflex
 - blocked by beta blockers
- ❑ **Nasal congestion**
- ❑ **Inhibition of ejaculation-** cause drug-induced impotence
 - Change the drug if the patient feels discomfort

β -Adrenergic Receptor Antagonists

- Based on their relative affinity for β_1 and β_2 receptors they are grouped into: -
 - None Selective β blockers: -
 - Propranolol, Carteolol, Penbutolol, Pindolol, Sotalol, Timolol, Nadolol, Bucindolol.
 - β_1 selective agents: -
 - Metoprololol, Acebutolol, Atenolol, Esmolol, Betaxolol, Celiprolol, Alprenolol, Bisoprolol, Nebivolol
 - Also referred to as cardio-selective beta blockers.

β -Adrenergic Receptor Blockers

Pharmacological Actions :

☐ CVS:

a. Heart (β_1):

- ↓ myocardial contraction
- ↓ HR
- ↓ AV-conduction & automaticity

☐ On Pulmonary System

- Bronchoconstriction (Nonselectives, propranolol)
 - Contraindicate in patients with asthma and COPD

Pharmacological Actions--

❑ CNS/Neurological:

Sedation (with propranolol, carvedilol

❑ Metabolic Effects

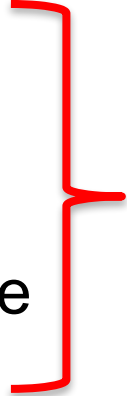
- Decrease glycogenolysis Hypoglycemia
- Inhibit sympathetic stimulation of lipolysis.reduce HDL cholesterol, increase LDL cholesterol

❑ On the eye: Reduce IOP in glaucoma ,decreased production of aqueous humor.

Therapeutic uses of β blockers

☐ Cardiovascular Diseases

- Hypertension
- Angina
- Congestive heart failure



Widely used

☐ Hyperthyroidism

- To reduce the peripheral manifestations of hyperthyroidism

☐ Treatment of Glaucoma

☐ Anxiety :

- ↓ sympathetic manifestations of anxiety (e.g., palpitations)

Adverse effects

❑ CVS:

- Bradycardia,
- hypotension, heart failure,
- AV block



β_1 blockers

❑ Bronchoconstriction, (β_2 blockers)

❑ Muscle pains & fatigue (β_2 blockers)

❑ Increase hypoglycemic effect of anti-diabetic drugs (β_2 blockers)

❑ Sleep disturbances, nightmares ($\uparrow$ with $\uparrow$ lipid soluble β blockers)

Contra-indications

- ❖ Heart failure (acute)
- ❖ Peripheral vascular disease
- ❖ Asthma
- ❖ Diabetes taking hypoglycemic drugs
- ❖ Hypothyroidism
- ❖ Combination with Ca-channel blockers

THANK YOU

